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Assessment Type: Group

Section 1

Problem Definition, Problem Statement and PDS

Note: attach your snapshot images

Problem Definition

Design Project: Camper Site Flooding Safety System

Project Background

Campsites located near rivers, lakes, or low-lying areas face flooding risks, especially during heavy rainfall. Flooding can damage camping equipment, disrupt activities, and endanger campers. To improve safety, innovative design solutions are needed that are practical, portable, and reliable in outdoor environments.

Project Task

The primary objective of this project is for you to design and develop a solution that enhances the safety of campers during flooding events. The system should either prevent or mitigate flooding risks at campsites, using a combination of mechanical engineering solutions, Internet of Things (IoT) technology, or a hybrid of both. The solution will need to be practical, cost-effective, and scalable for real-world use.

Problem Statement

The primary issue addressing this project is the critical absence of real-time, on-site monitoring systems at campsites. Current flood alert mechanisms operate on a macro scale, providing broad regional warnings that lack the granularity required for specific campsite safety. This lack of localized data creates a significant notification delay, often referred to as latency, between the onset of rising water levels and the reception of a warning. Consequently, campers are left with insufficient time to react, rendering them unable to evacuate before water levels become hazardous. This delay significantly increases the risk of entrapment, property damage, and casualty during flash flood events.

To mitigate these risks, the proposed solution is the "AQUASS" system—a rugged IoT-based device designed specifically for continuous lakeside monitoring. The system is engineered to perform real-time water level analysis and autonomously execute a decision-making process based on safety thresholds. The core design philosophy focuses on a dual-alert mechanism that integrates immediate local sirens for on-site campers and remote Blynk notifications for off-site operators or authorities. This approach ensures reliable, all-weather operation and provides a redundant warning system to guarantee that alerts are received regardless of the campers' immediate attentiveness.

Product Design Specifications (PDS)

Category	Specification	Description / Target / Rationale
Function	Flood detection and early warning	Detects rising water levels near campsites and triggers alerts for evacuation.
Application Area	Campsite or recreational sites near lakes	Fixed installation near potential flood-prone zones.
Performance requirements	Detection range	Detects water-level rise within 0–200 cm range with $\pm 2\%$ accuracy.
	Response time	Alert within ≤ 5 seconds after the danger threshold exceeded.
	Alert coverage	Local alert (LED, buzzer) audible within 30 m radius.
Power Requirement	Supply	12V DC (solar + Li-ion battery backup).
	Operation time	Minimum 7 days continuous operation without sunlight.
	Power consumption	< 1.5 W average (low-power IoT mode).
Dimensions	Pole height	$\geq 1\text{m}$
	Casing	Length ≤ 50 cm Width ≤ 50 cm Height ≤ 50 cm
Communication / Connectivity	Local and remote alerts	GSM or LoRa for off-grid sites; Wi-Fi optional where available.
	IoT integration	Blynk / ThingSpeak for data monitoring and notifications.

	Materials and Construction	Enclosure	Weatherproof IP65+ ABS/polycarbonate enclosure.
		Mounting	Fixed installation on pole or platform near riverbank.
		Environmental protection	UV-resistant, waterproof, and corrosion-proof materials.
	Safety Requirements	Electrical safety	All low-voltage operation ($\leq 12\text{V DC}$).
		Environmental safety	Non-toxic, sealed components; no hazardous emissions.
		Human safety	Visual and audible alerts comply with campsite noise limits ($< 80\text{ dB}$).
		Structural safety	Safety factor of the most critical point of the structure > 3.5
	Environmental Conditions	Operating temperature	0°C to 60°C .
		Humidity	10–95% RH, non-condensing.
		Weather resistance	Must function during heavy rain, humidity, and minor debris exposure.
	Reliability and Maintenance	Reliability	System uptime $\geq 95\%$ during continuous monitoring.
		Maintenance	Annual battery inspection; sensor calibration every 6 months.
		MTBF (Mean Time Between Failures)	≥ 5000 hours.
	Ergonomics and Usability	User interface	Simple visual indicator (LED status) and mobile notification.
		Installation	Quick installation (≤ 6 hours) with minimal tools and instructions.
		Accessibility	Accessible maintenance port with waterproof seals.
	Aesthetic / Design Factors	Design form	Compact and have eye-catching colour such as white.
		Visibility	LED indicators visible in daylight and night.

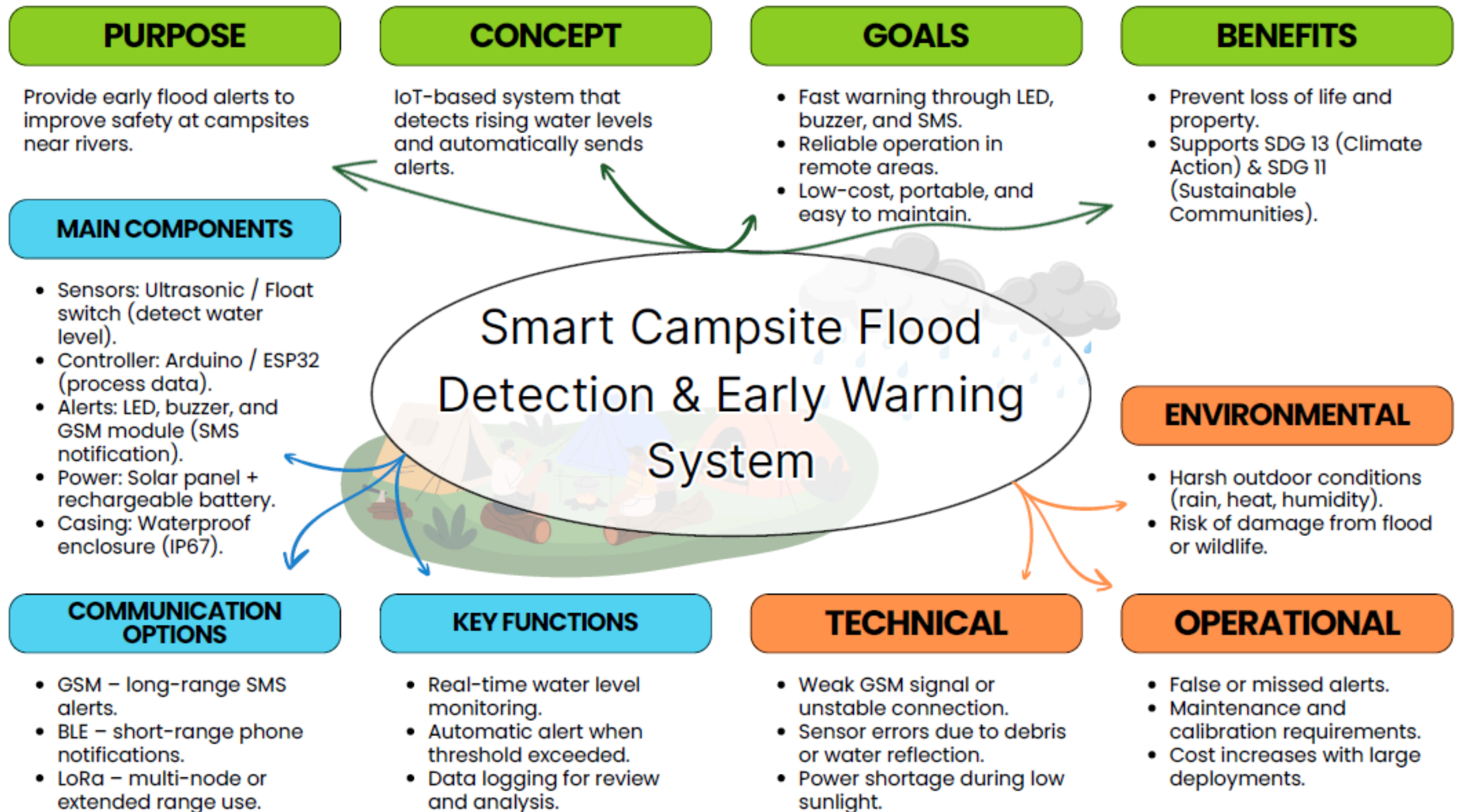
Standards / Compliance	Electrical & communication	Compliant with CE, FCC IoT, and IP65 standards.
	SDG alignment	Supports SDG 13 (Climate Action) and SDG 11 (Sustainable Communities).
Cost Target	Prototype cost	≤ RM 400 per unit (excluding solar panel).
	Large-scale deployment	Target ≤ RM 500 per unit in production (excluding microcontroller / microprocessor, wirings).
Constraints	Location constraint	Installed within 1m from the water body for accurate monitoring.
	Power source constraint	Dependent on solar exposure; must ensure energy balance.
	Communication constraint	GSM coverage required if LoRa is not available.
Testing and Validation	Validation methods	Simulated flood tank testing, field trial at real campsite.
	Key performance indicators	Detection accuracy, alert reliability, power endurance, communication stability.

Assessment Type: Group

Section 2

Data gathering (mind maps or flow chart)

Note: attach your snapshot images

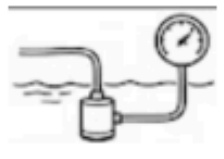
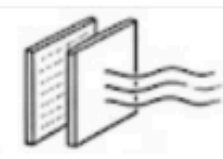
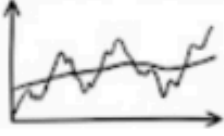
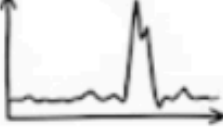
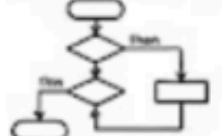
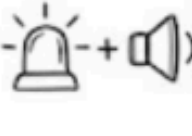
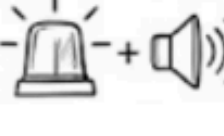
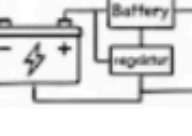
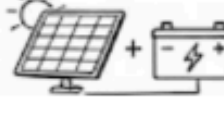


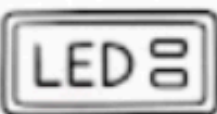
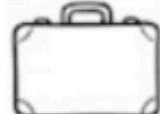
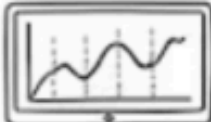
Assessment Type: Group

Section 3

Morphological chart

Note: attach your snapshot images

Main Function	Option 1	Option 2	Option 3	Option 4	Option 5
1. Detect water level changes	Threshold-based contact detection (simple trigger logic) 	Ultrasonic distance measurement using time-of-flight 	Hydrostatic pressure-derived level estimation 	Capacitive sensing of surface proximity 	Water level rising rate analysis from time-series data 
2. Process and classify data	Fixed water level limits (3 stages) 	Changing water level limits based on the current water trend 	Average reading to avoid false triggering 	Instant reaction to sudden change 	Rule-based decision (if-then logic) 
3. Communicate data and alerts	Local LED/buzzer indication only 	GSM/SMS message to registered phone 	IoT app alert via Blynk/MQTT 	LoRaWAN long-range gateway 	Web dashboard or Telegram bot for remote monitoring 
4. Provide early warning (alert strategy)	Multi-color LED for level indication 	Buzzer alarm with increasing tone 	Combined audio + visual alert 	Push notification + on-site alarm 	Voice/siren broadcast for evacuation signal 
5. Manage power supply and efficiency	Fixed battery power with regulator 	Solar charging with energy controller 	Hybrid solar + battery auto-switch 	Duty-cycled low-power firmware 	Smart power scheduling based on ambient light 

6. Components protection	IP65 waterproof housing 	Elevated mounting above expected flood line 	Enclosure with drain and vent 	Redundant sensing (dual-sensor safety) 	Self-diagnostic system to check faults 
7. User interaction & maintenance	Simple push-button calibration and reset 	Smartphone interface for manual control 	LED status for battery/sensor health 	Text message status query command 	Scheduled maintenance alert via app 
8. Mounting / Installation Method	Fixed pole mount near water edge 	Clamp-on installation at campsite fence or post 	Floating platform anchored to lakeside 	Wall-mount box near control area 	Tripod stand or ground stake for flexible setup 
9. System deployment approach	Permanent installation by campsite operator 	Semi-permanent system for seasonal use 	Portable kit for mobile campsite deployment 	Shared community unit between nearby campsites 	Modular units that can be extended with more sensors 
10. Data recording and analysis	Local SD card data logging 	Real-time dashboard visualization 	Periodic report generation via cloud 	Trend analysis for water level prediction 	AI-assisted data correlation for risk mapping 

Assessment Type: Individual

Section 4

Concept Generation, and Concept Redevelopment

Each group member must submit their individual contribution for this section. Please ensure your name is clearly written on all submitted materials.

Note:

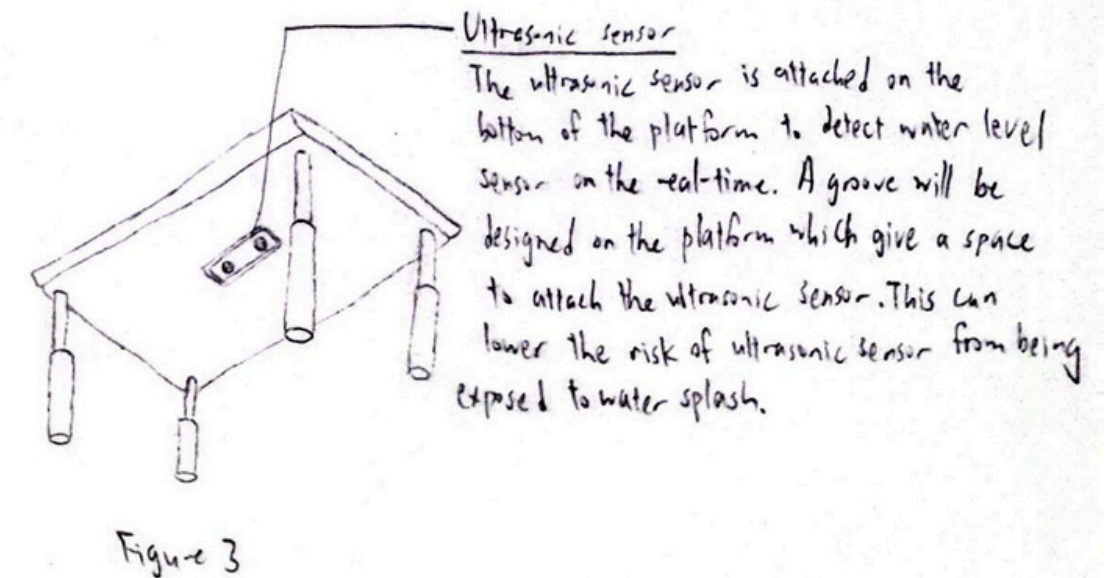
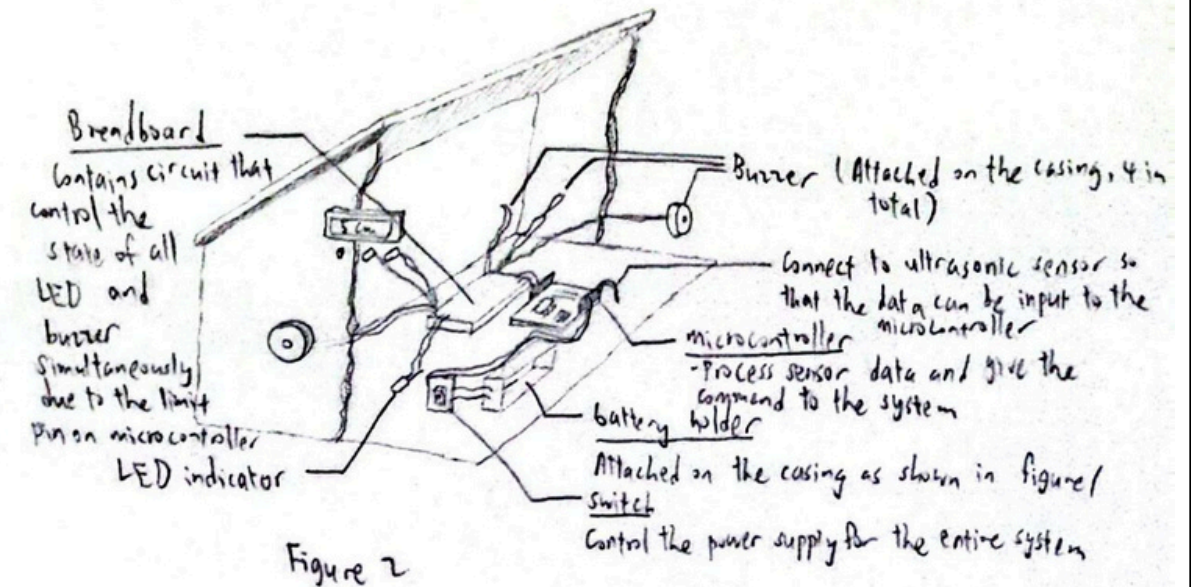
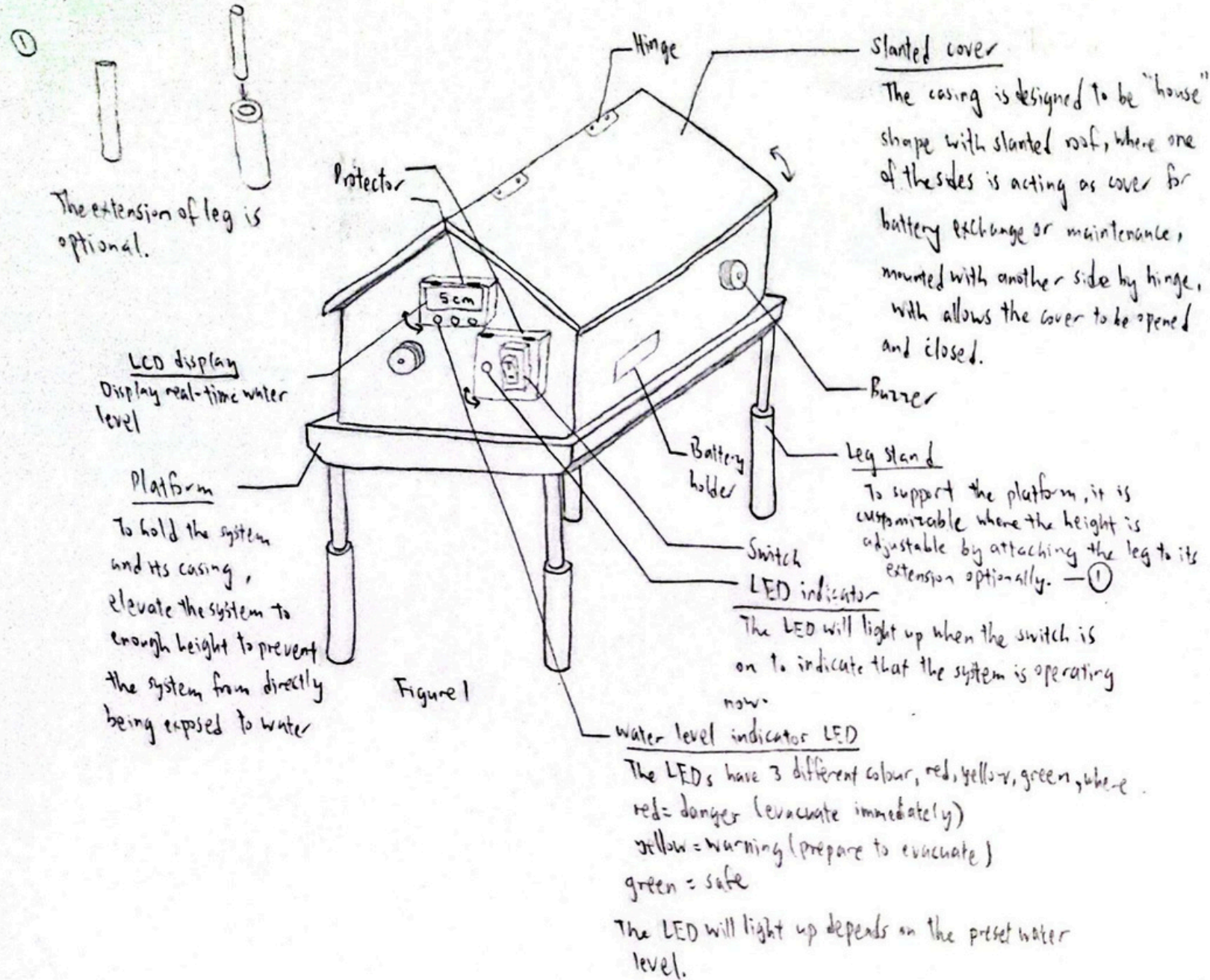
Attach your snapshot images of:

- 3 minimum sketches of Concept Generation
- and
- 1 sketch of Concept Redevelopment

Concept Generation

NAME: HEW YEE HANG
GROUP: 7

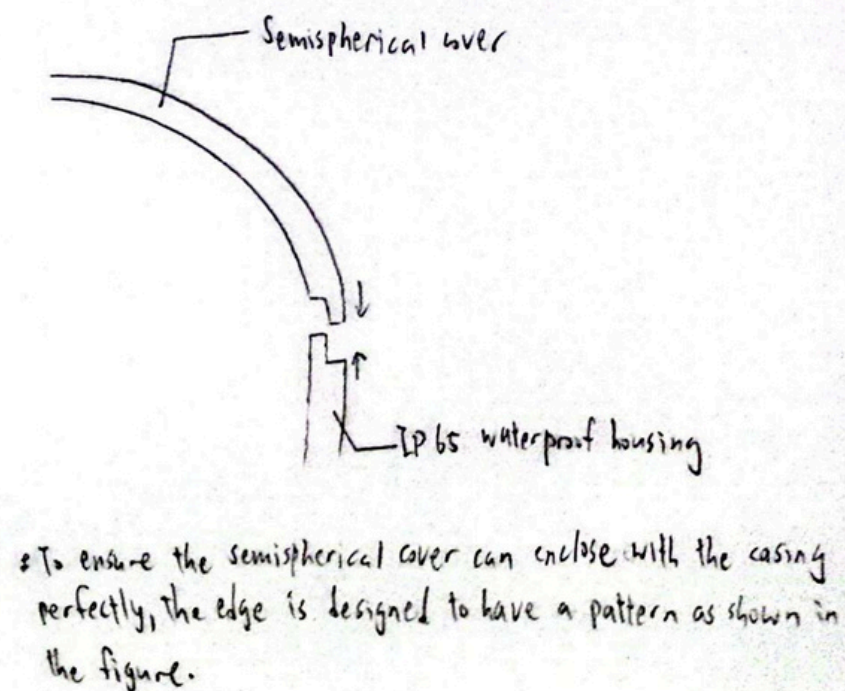
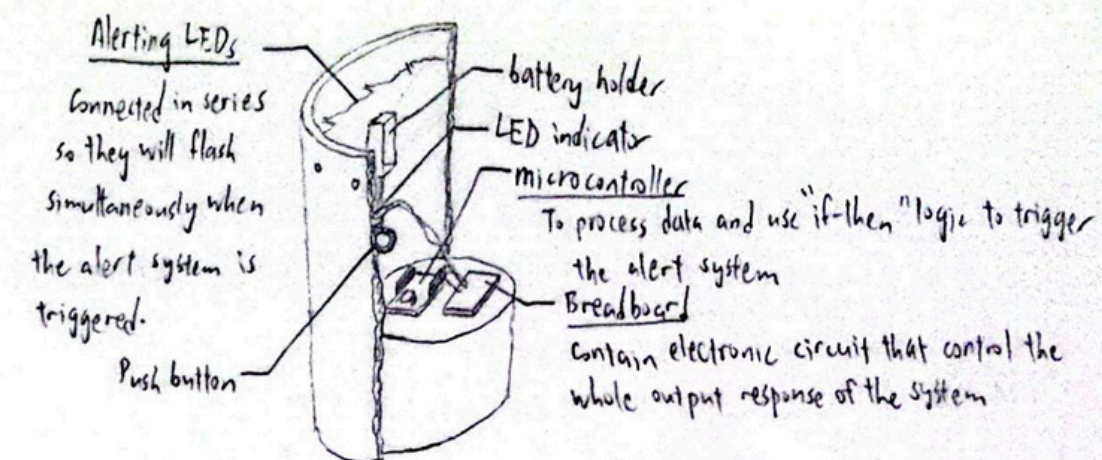
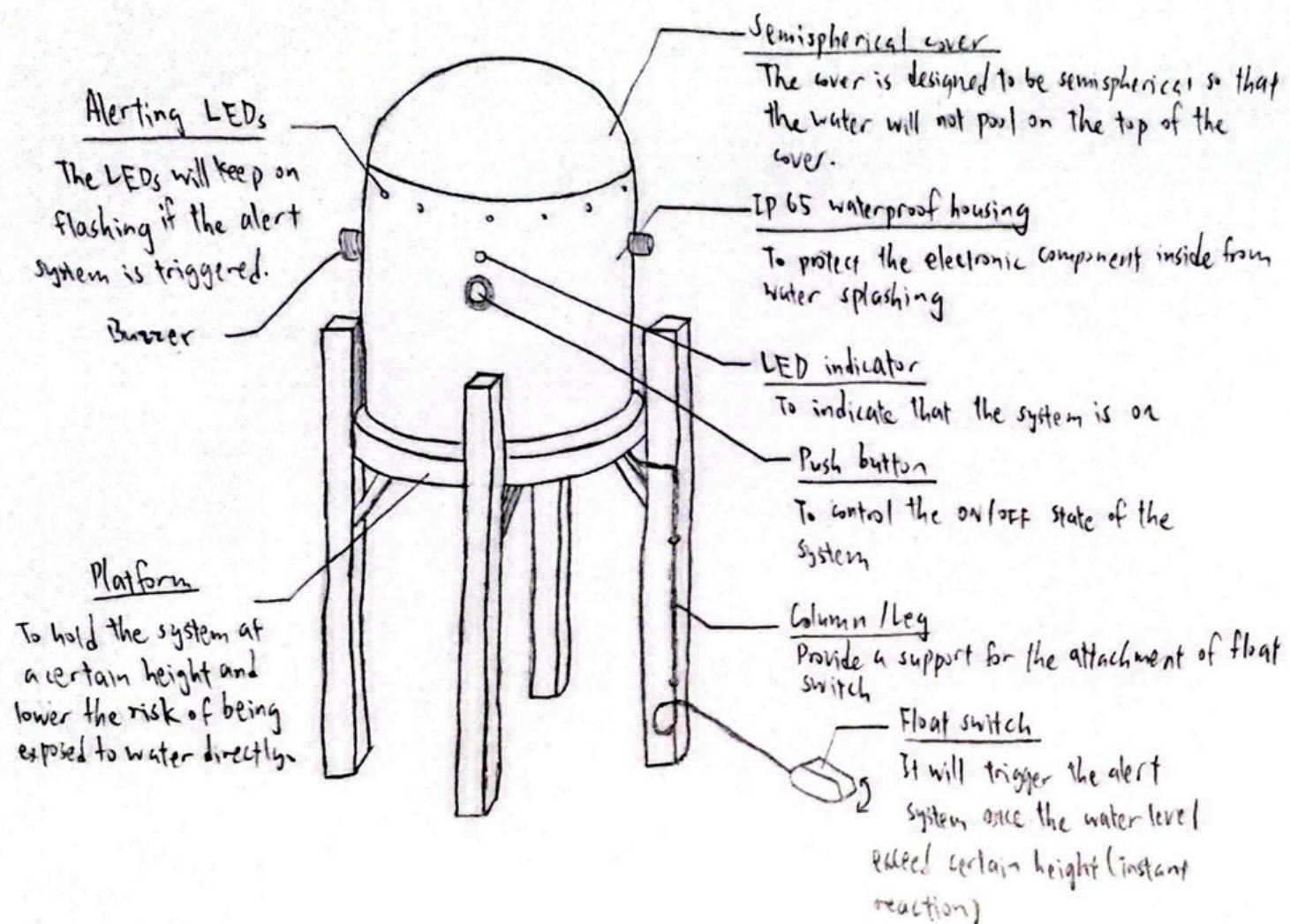
Conceptual Sketching 1



NAME: HEW YEE HANG

GROUP: 7

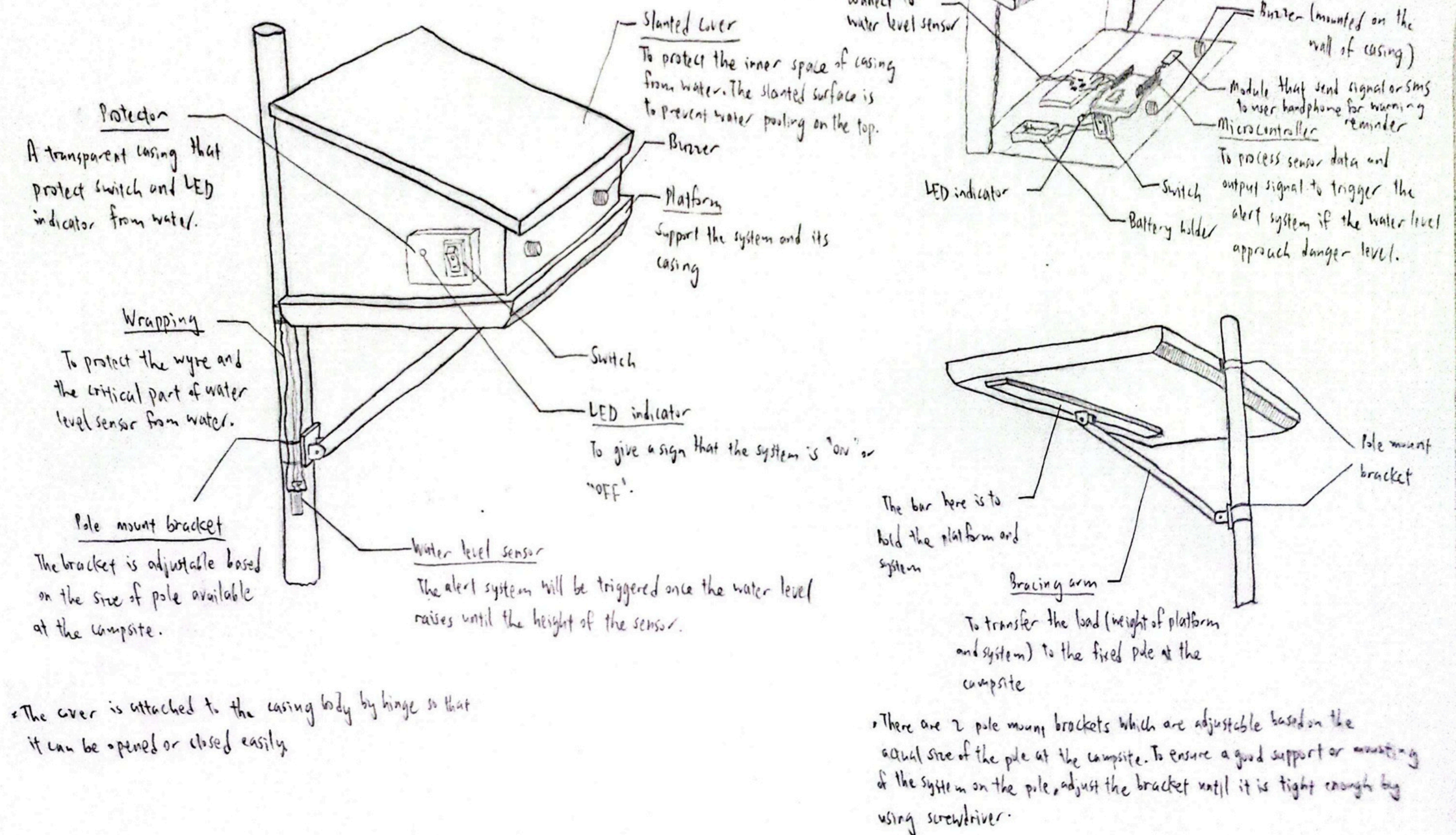
Conceptual Sketching 2



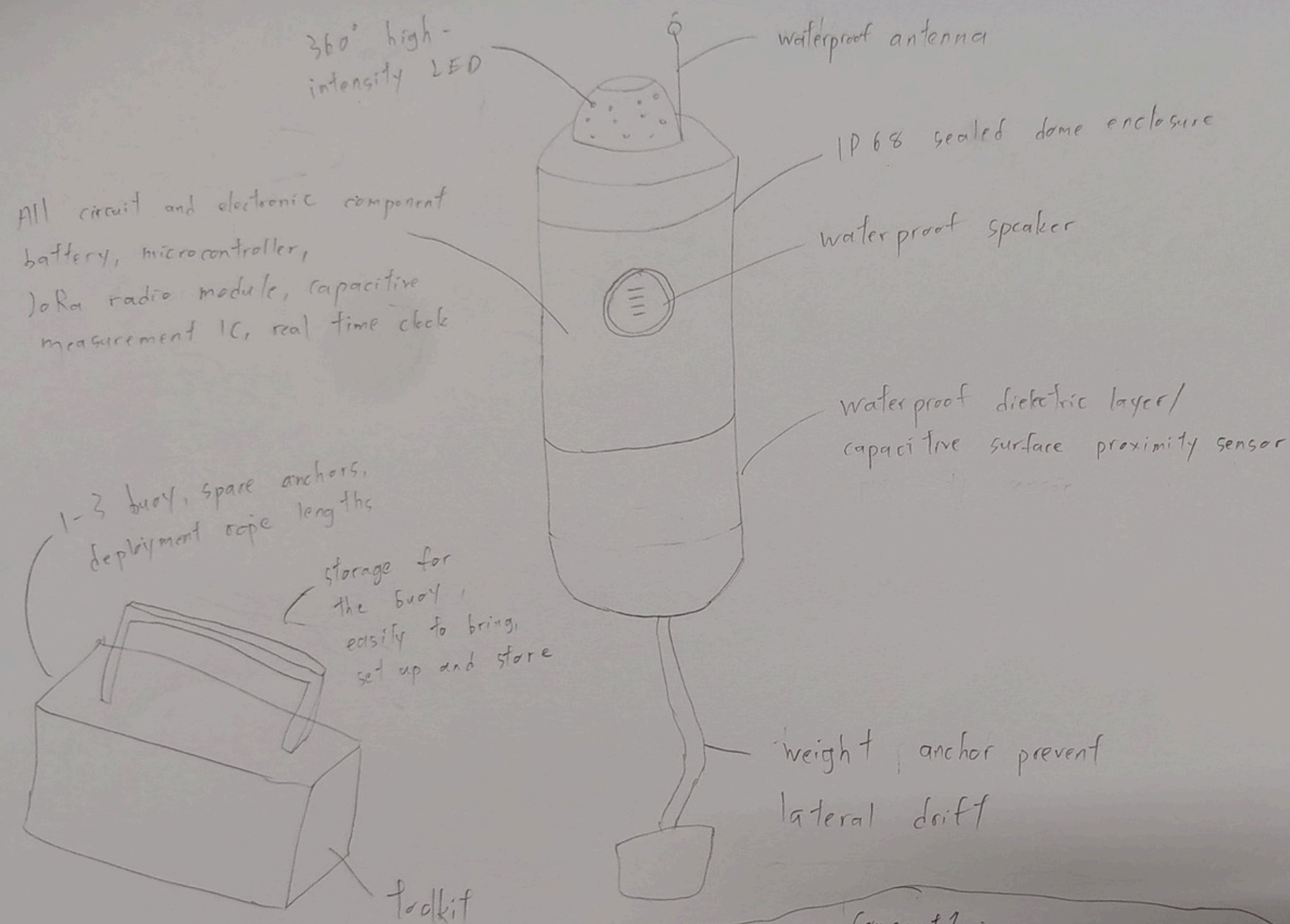
NAME: HEW YEE HANG

GROUP: 7

Conceptual Sketching 3



Hew Yee Hang Conceptual sketching 3

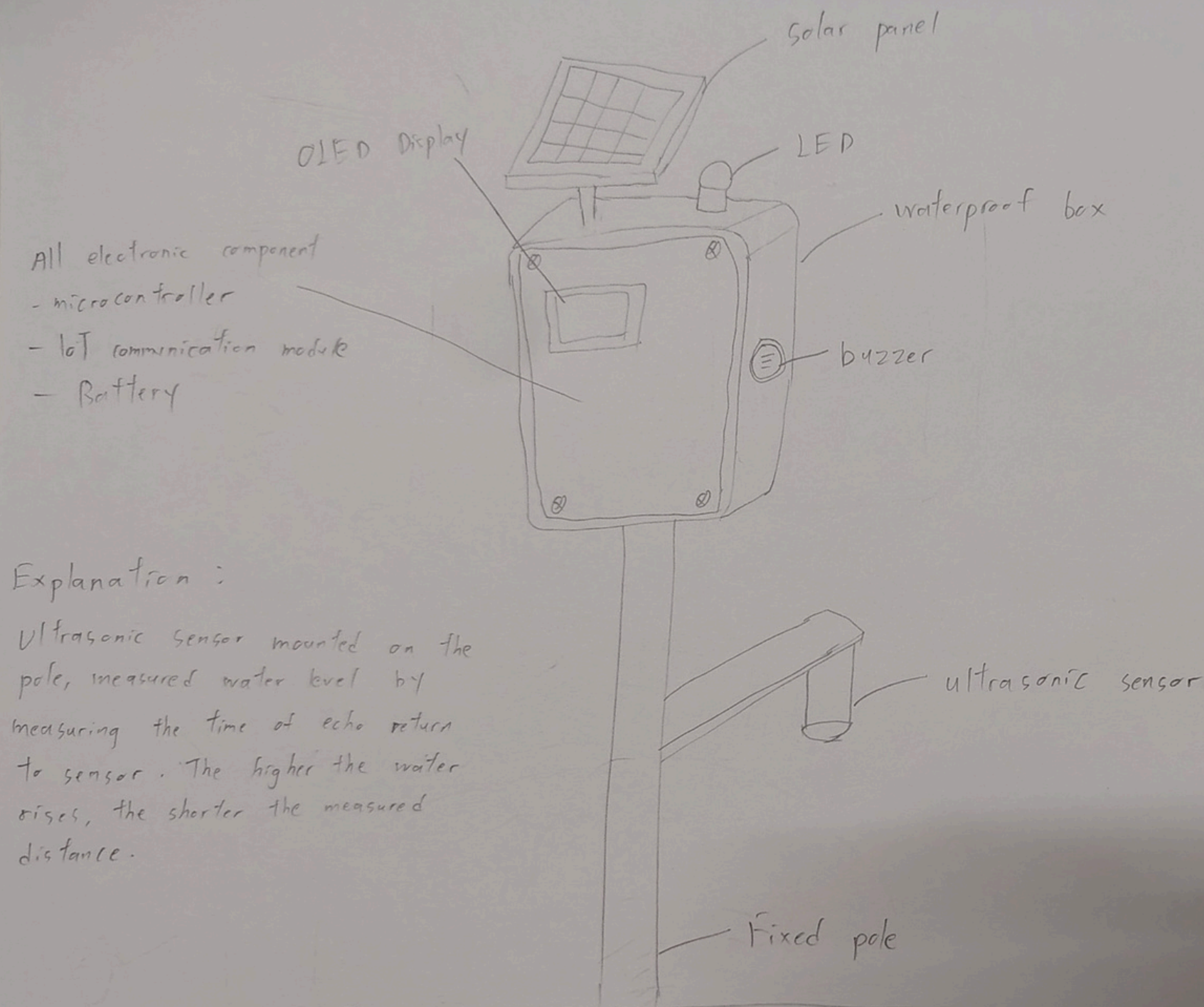


Concept 1:
A self-contained floating sensor that rises with the water and provides instant proximity-based flood alert

Explanation:

The buoy senses water level is when the higher the water rises, the more of its body gets submerged, and that submerged section strongly changes the capacitive sensor's electric field

- ① Capacitive sensing of surface proximity
- ② Rule-based decision
- ③ LoRaWAN long-range gateway
- ④ Combined audio + visual alert
- ⑤ Fixed battery power with regulator
- ⑥ IP65 waterproof housing
- ⑦ Scheduled maintenance alert via app
- ⑧ Floating platform anchored to riverbank
- ⑨ Portable kit for mobile campsite deployment
- ⑩ Periodic report generation via cloud



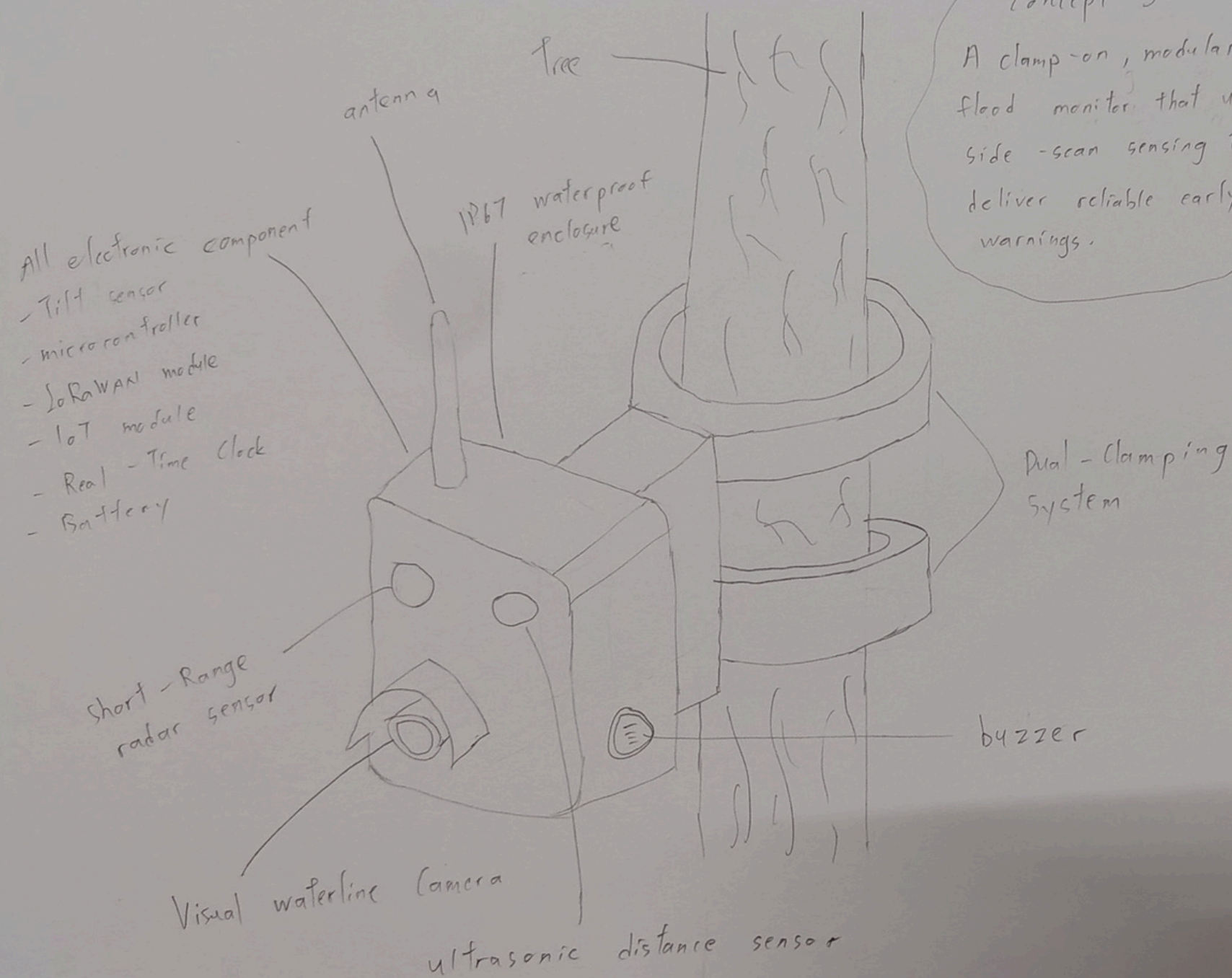
Explanation :

Ultrasonic sensor mounted on the pole, measured water level by measuring the time of echo return to sensor. The higher the water rises, the shorter the measured distance.

Concept 2 :

A pole-mounted ultrasonic station that continuously monitors water level and issues solar-powered IoT alerts.

- ① Ultrasonic distance measurement using time of flight
- ② Rule-based decision
- ③ IoT app alert via Blynk / MATT
- ④ Voice / siren broadcast for evacuation signal
- ⑤ Hybrid solar + battery auto-switch
- ⑥ Self-diagnostic system to check faults
- ⑦ Smartphone interface for manual control
- ⑧ Fixed pole mount near water edge
- ⑨ Permanent installation by campsite operator
- ⑩ Real-time dashboard visualization



Explanation:

The device measures water level by using a side facing radar sensor to detect the distance to the water surface, then calculates the actual water height based on its known mounting height and sensor angle.

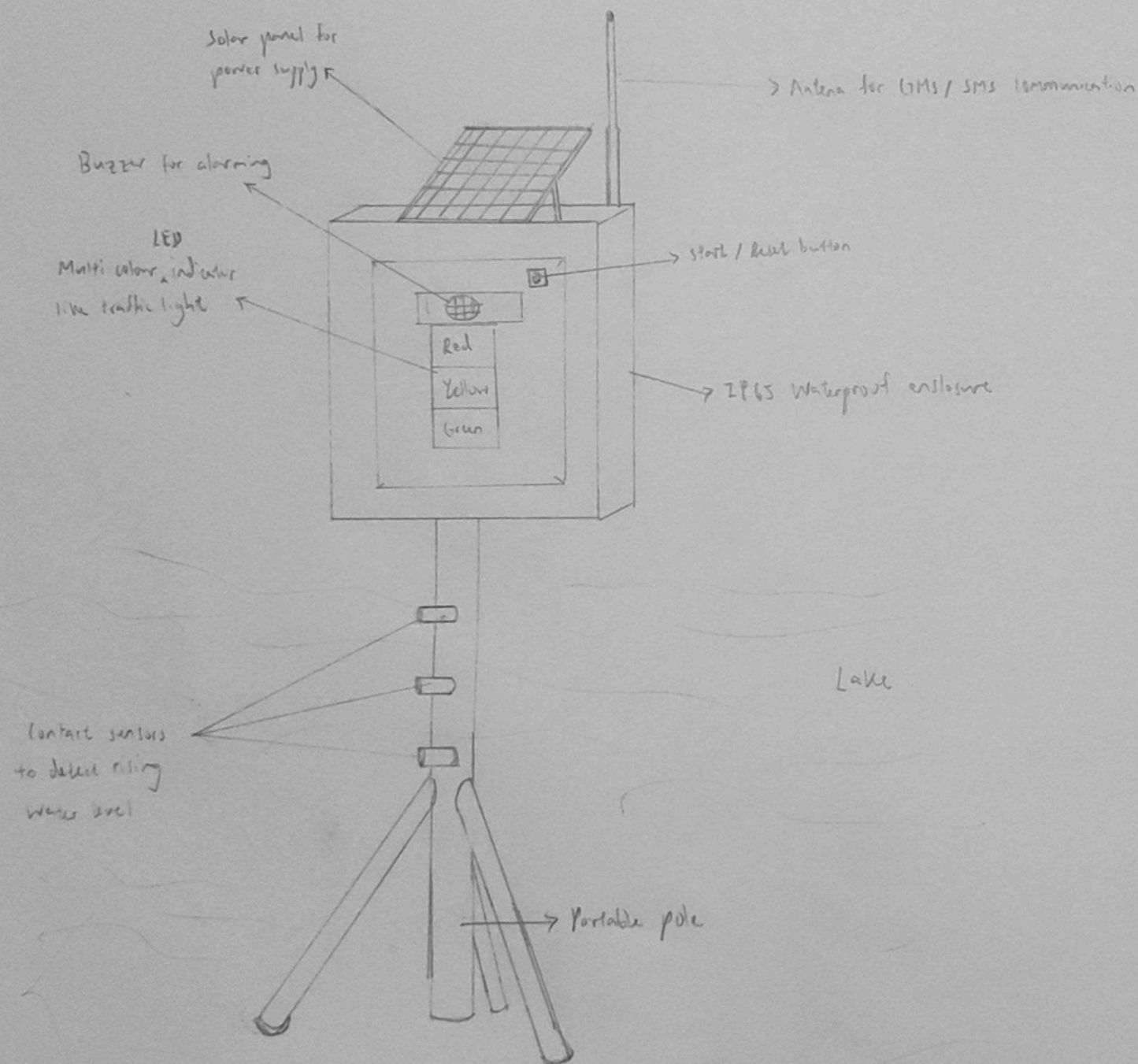
- ① Water level rising rate analysis from time series data
- ② Rule-based decision
- ③ LoRaWAN long-range gateway
- ④ Push notification + on-site alarm
- ⑤ Fixed battery power with regulator
- ⑥ IP65 waterproof housing
- ⑦ Smartphone interface for manual control
- ⑧ Clamp-on installation at rampsite fence or post
- ⑨ Portable kit for mobile rampsite deployment
- ⑩ Trend analysis for water level prediction

Name: Lawrance Sim Lip We

Group: 7

Conceptual Sketching 1

→ This system is design for simplicity and local warning at a fixed composite.



Main Functions

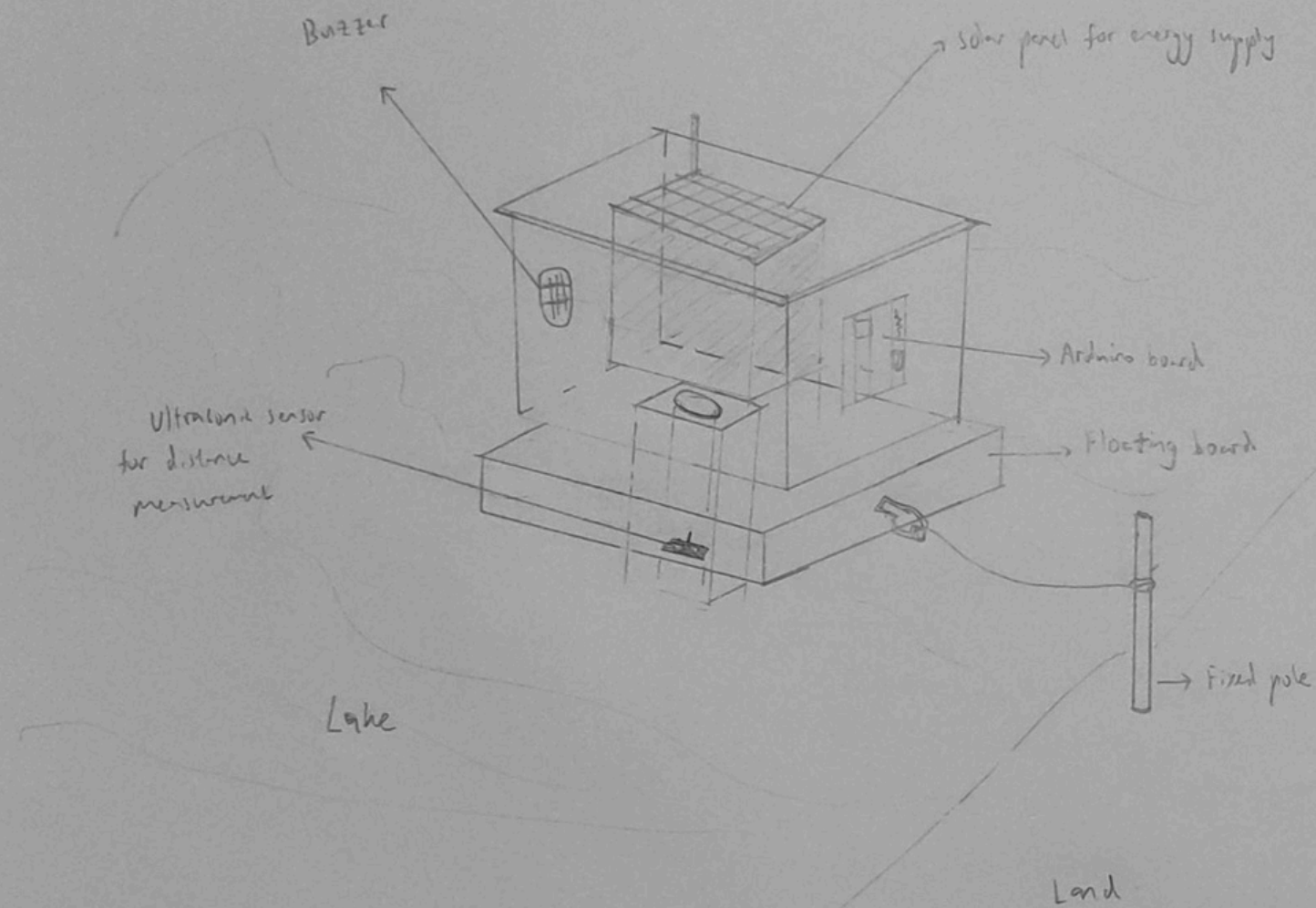
1. Detect water level changes : Threshold based contact detection (simple trigger logic)
2. Process and classify data : Fixed water level limit (3 stages)
3. Communicate data and alerts : Local LED / buzzer indication only.
4. Provide early warning : Multi colour LED for level indication.
5. Manage power supply and efficiency : solar charging with energy controller
6. Component protection : IP65 waterproof housing
7. User interaction & maintenance : Simple push button calibration and reset
8. Mounting / Installation method : Fixed pole mount near water edge
9. System Deployment approach : Semi-permanent system for seasonal uses.
10. Data recording & analysis : Local SD card data logging

Name: Lawrance Sim Lip We

Group: 7

Conceptual sketching 2

→ This design is on high tech remote with non-contact ultrasonic sensor, a mobile IoT app for alerts and a floating platform for dynamic water level.



- ① Ultrasonic distance measurement using time-of-flight
- ② Rule-based decision
- ③ IoT app alert via Blynk / MQTT
- ④ Push notification + on-site alarm
- ⑤ Solar charging with energy controller
- ⑥ Elevated mounting above expected floods
- ⑦ LED status for battery / sensor health
- ⑧ Floating platform anchored to lakeside
- ⑨ Permanent installation by lakeside operator
- ⑩ Periodic report generation via cloud

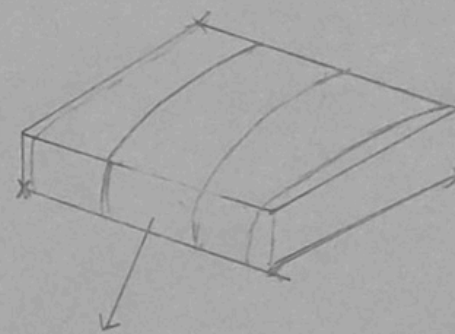
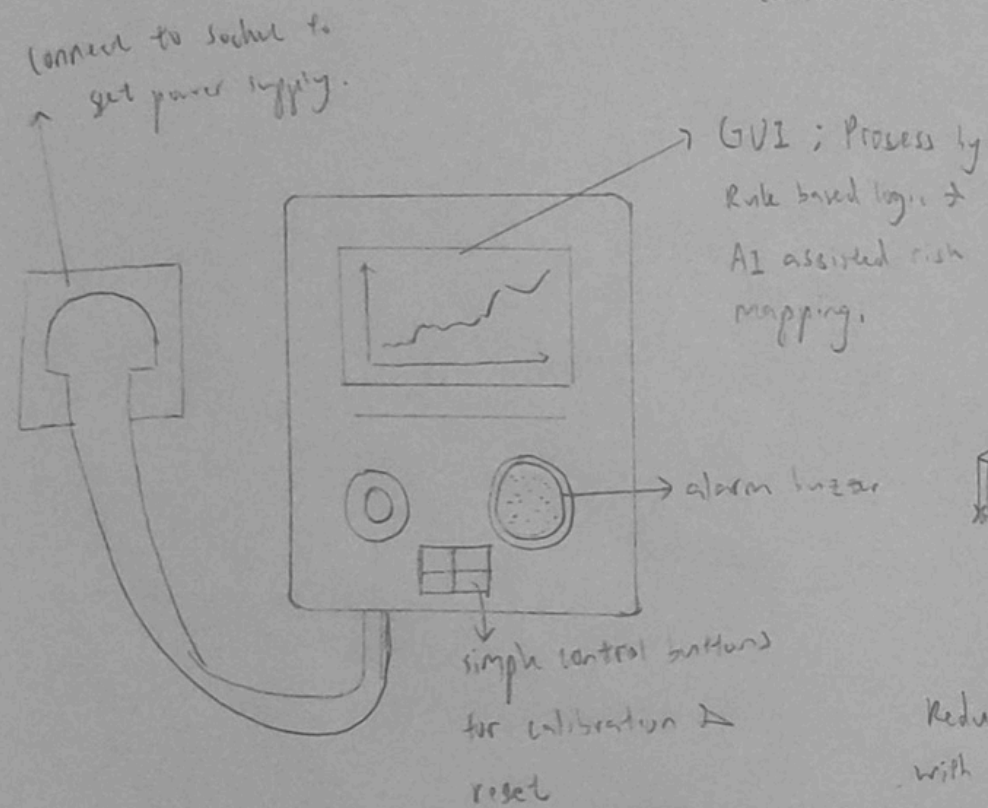
Lawrance Sim Lip We Conceptual sketching 2

Name: Lawrance Sim Lip We

Group: 7

Conceptual sketching 3

→ This design is future-oriented, focusing on preventive action through advanced data interpretation and AI assisted risk mapping, utilising dual redundant sensors and a wall-mounted control box.

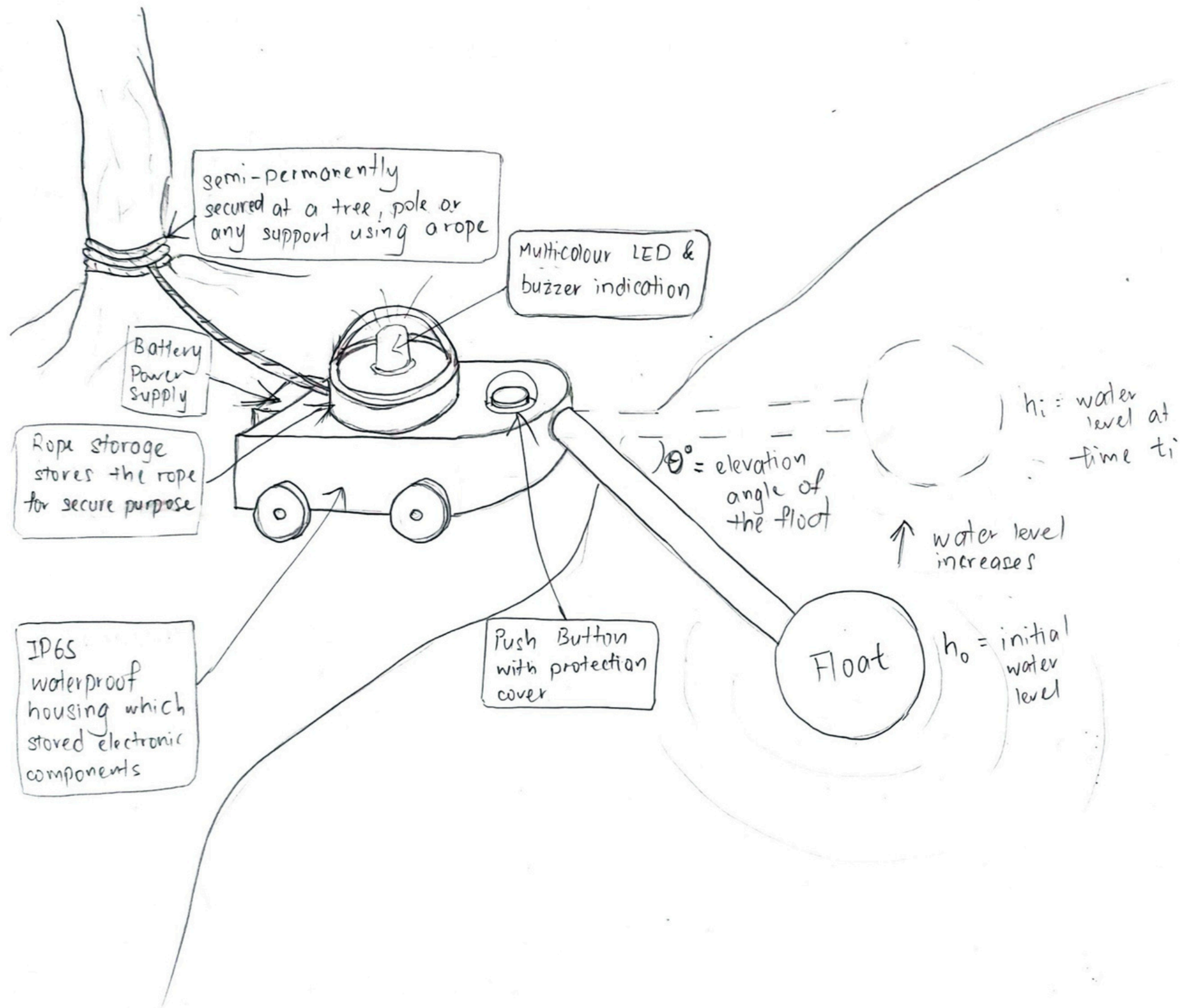


Redundant sensing (dual sensor safety) with two sensor inside the cage that fixed inside the lake. Both the sensor will be hydrostatic pressure-derived level estimation sensors. Only if both sensor trigger alarm, then will detect as flood occurrence.

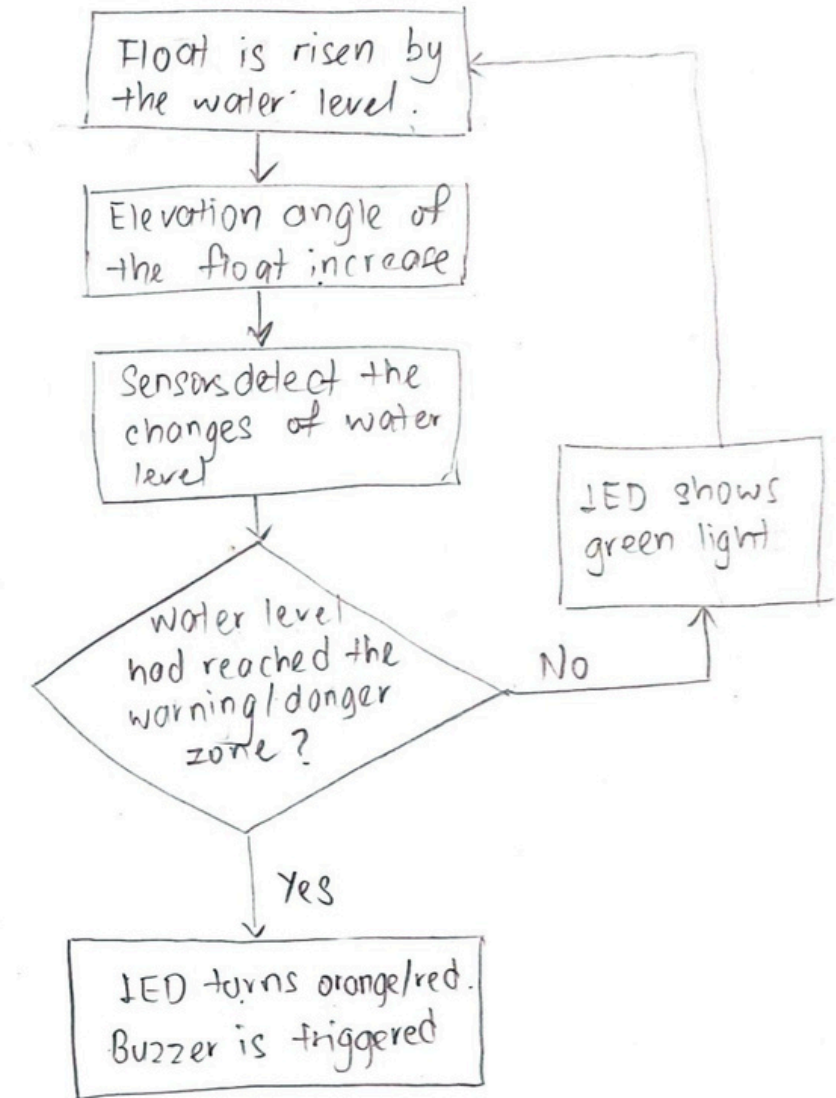
- ① Hydrostatic pressure-derived level estimation
- ② Average reading to avoid false triggering
- ③ Web dashboard for remote monitoring
- ④ Push notification & on-site alarm
- ⑤ Fixed battery power with regulator
- ⑥ IP65 waterproof housing
- ⑦ Simple push-button calibration and reset
- ⑧ Fixed inside the lake
- ⑨ Modular unit that can be extended with more sensors.
- ⑩ Trend analysis for water level prediction

Liew Shan Jie A22EM0107

Sketch 1

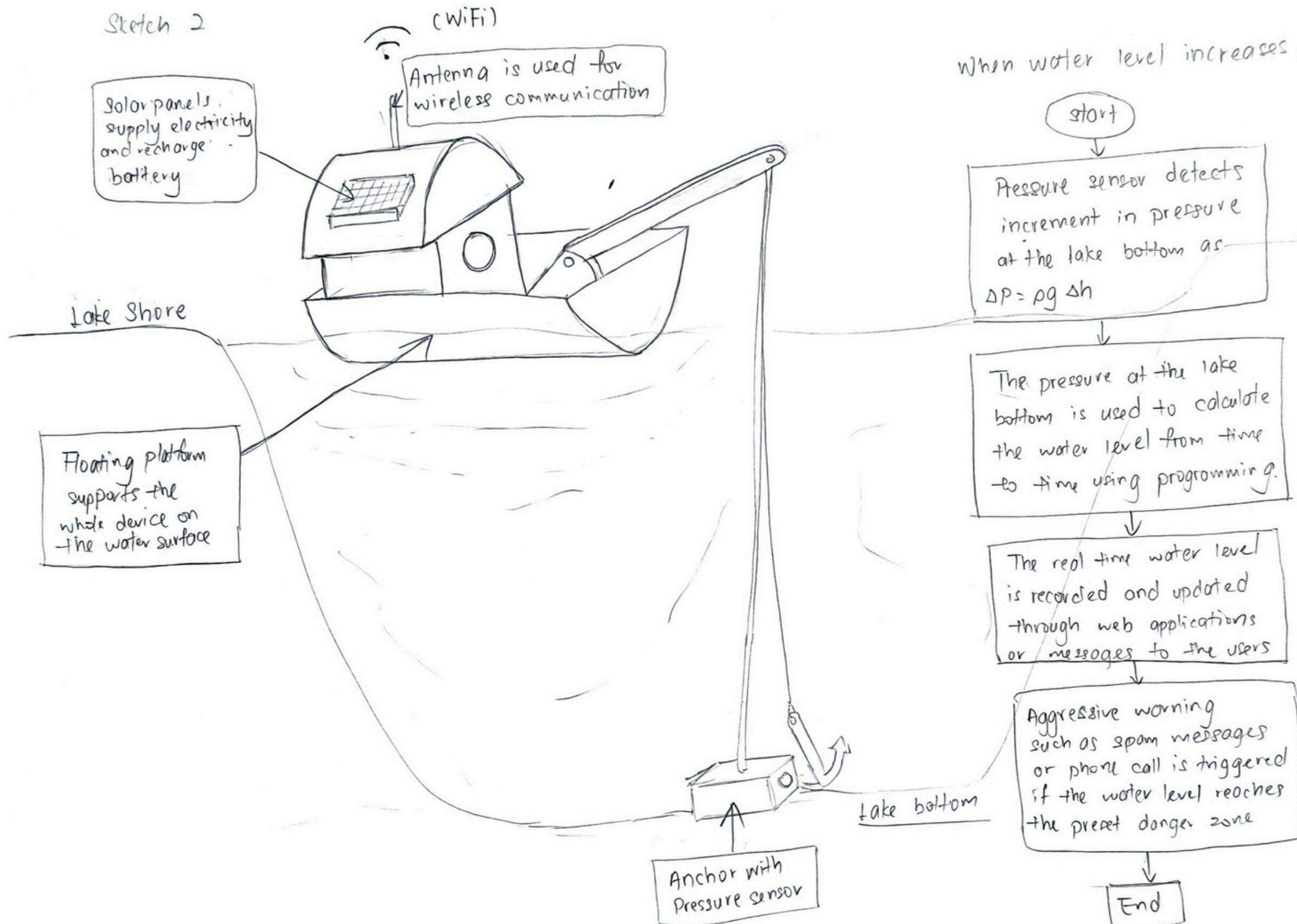


when water level is increasing,



Liew Shan Jie A22EM0107

Sketch 2

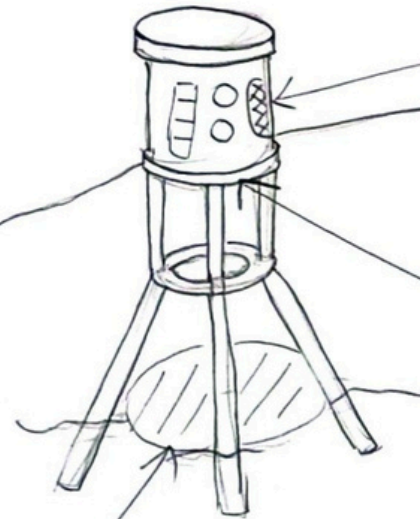


Liew Shan Jie Conceptual sketching 2

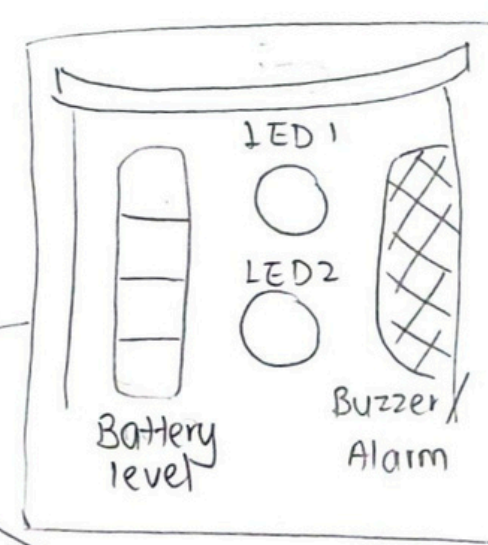
Liew Shan Jie A02EM0107

Sketch 3

- Portable
- Small
- Light-weight

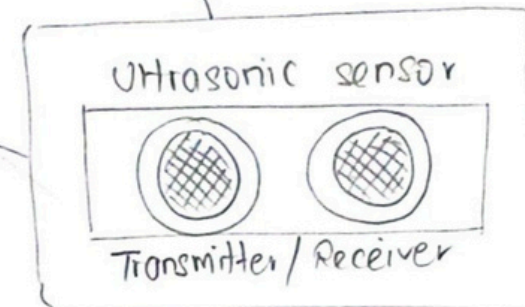


Ultrasonic sensor has low accuracy towards the water surface, hence a floating platform with good reflection of ultrasonic wave is secured within the tripod legs



LED 1 { Green : safe
Red : Danger

LED 2 { Yellow light : charging
Yellow blink : battery $\leq 50\%$



Start
Floating platform has a variable height depending on the water level.

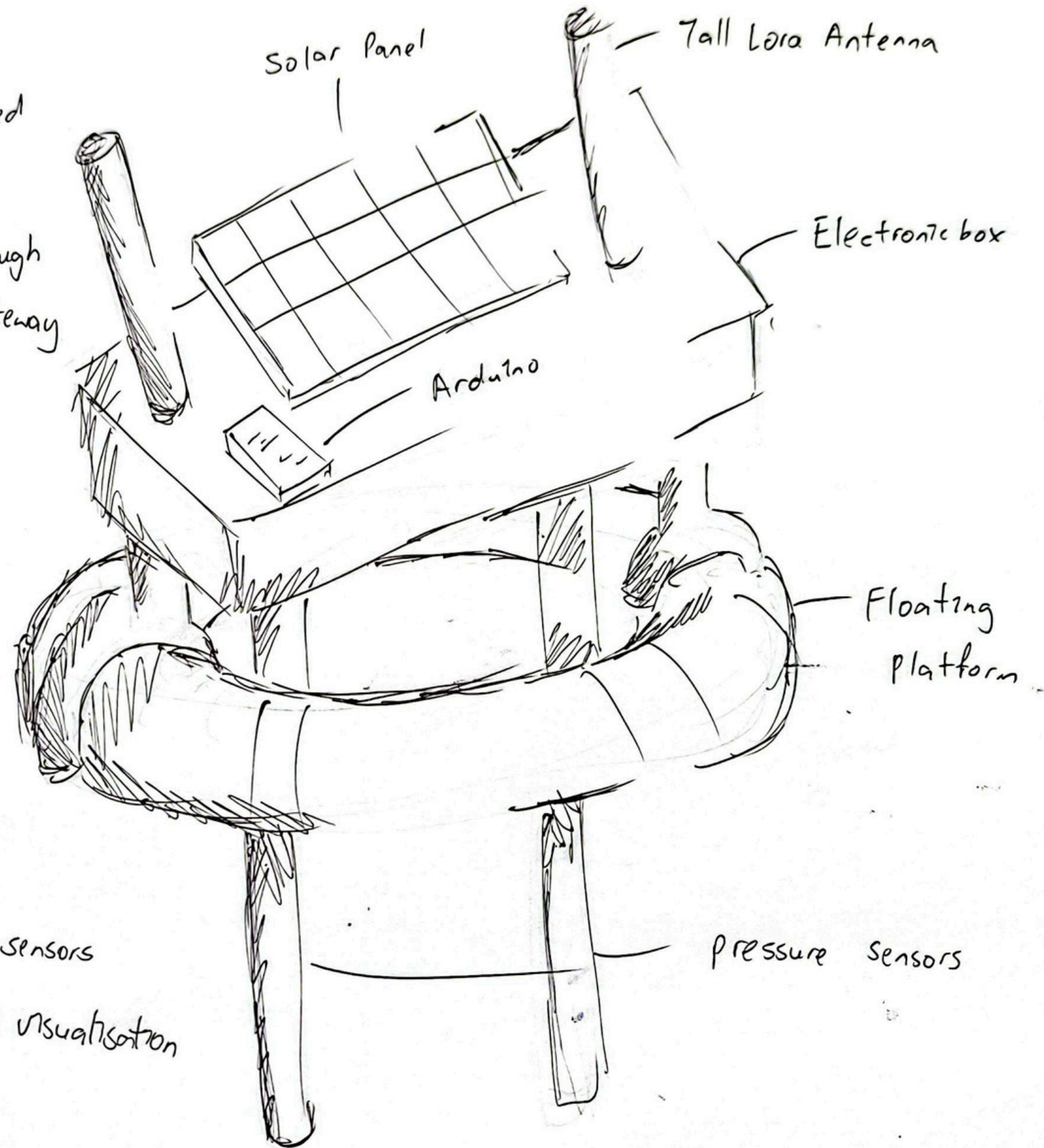
Ultrasonic sensor detects the distance of the floating platform from time to time

The water level rising rate is calculated using programming

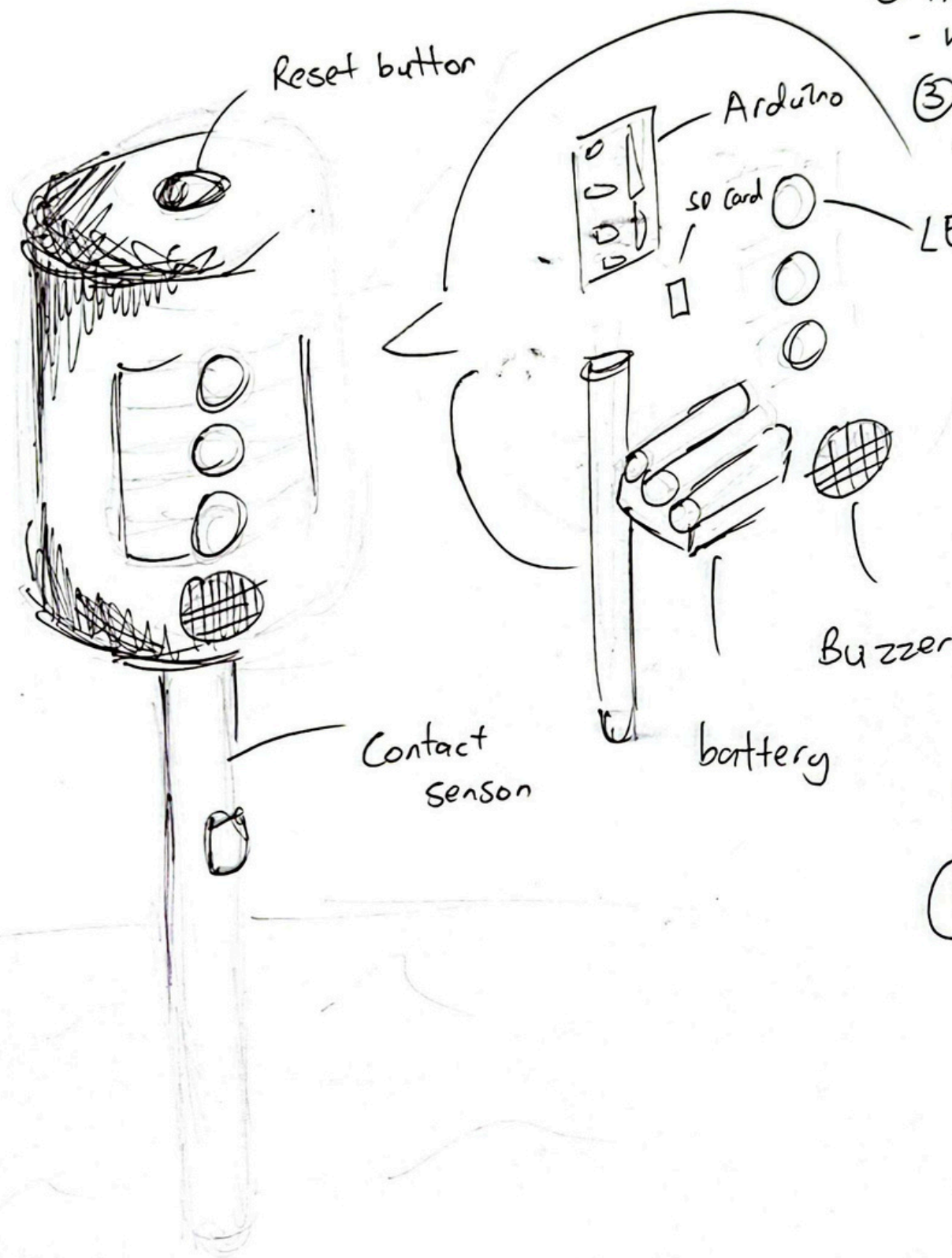
Warning and buzzer are triggered if the water level rising rate is faster than the preset safety value or the water level distance detected is higher than the safety limit.

End

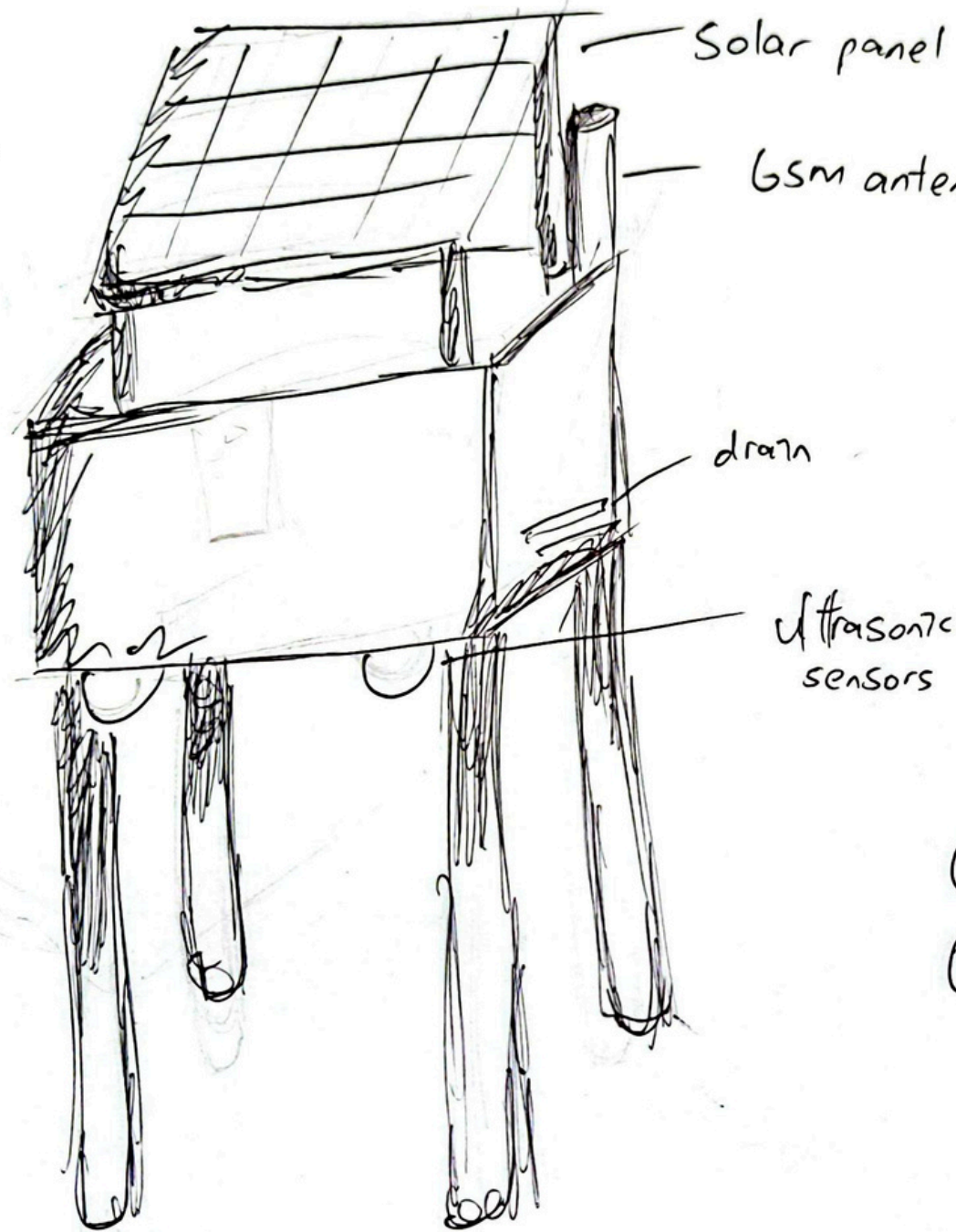
- ① Detect water
 - Hydrostatic pressure based
- ② Process & classify data
- ③ Communicate alerts through LoRaWAN long-range gateway
- ④ Early warning
 - Push notification + alarm
- ⑤ Solar powered
- ⑥ Dual sensor
- ⑦ Smart phone interface
- ⑧ Floating platform anchored to riverbank
- ⑨ Scalable with more sensors
- ⑩ Real time dashboard visualisation



Pang Zi Hao Conceptual sketching 1



- ① Detect water level
- Threshold based contact sensor
- ② Process data
- water level 17m7+
- ③ Communicate alerts
- Local LED and buzzer
- ④ Warning
- LED for level indication
- ⑤ Power
- battery powered
- ⑥ IP65 waterproof
- ⑦ Push button reset
- ⑧ Mounted near water edge
- ⑨ Install easily
- ⑩ Local SD card data logging



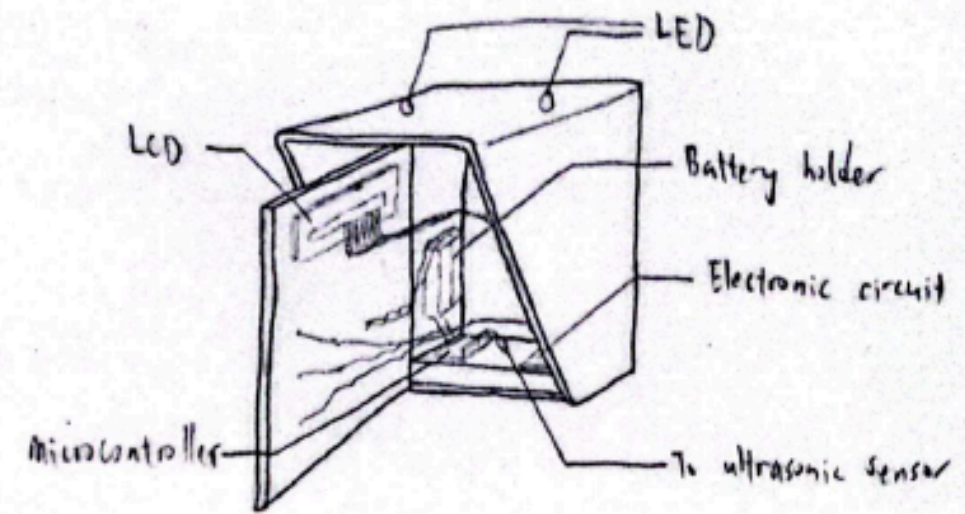
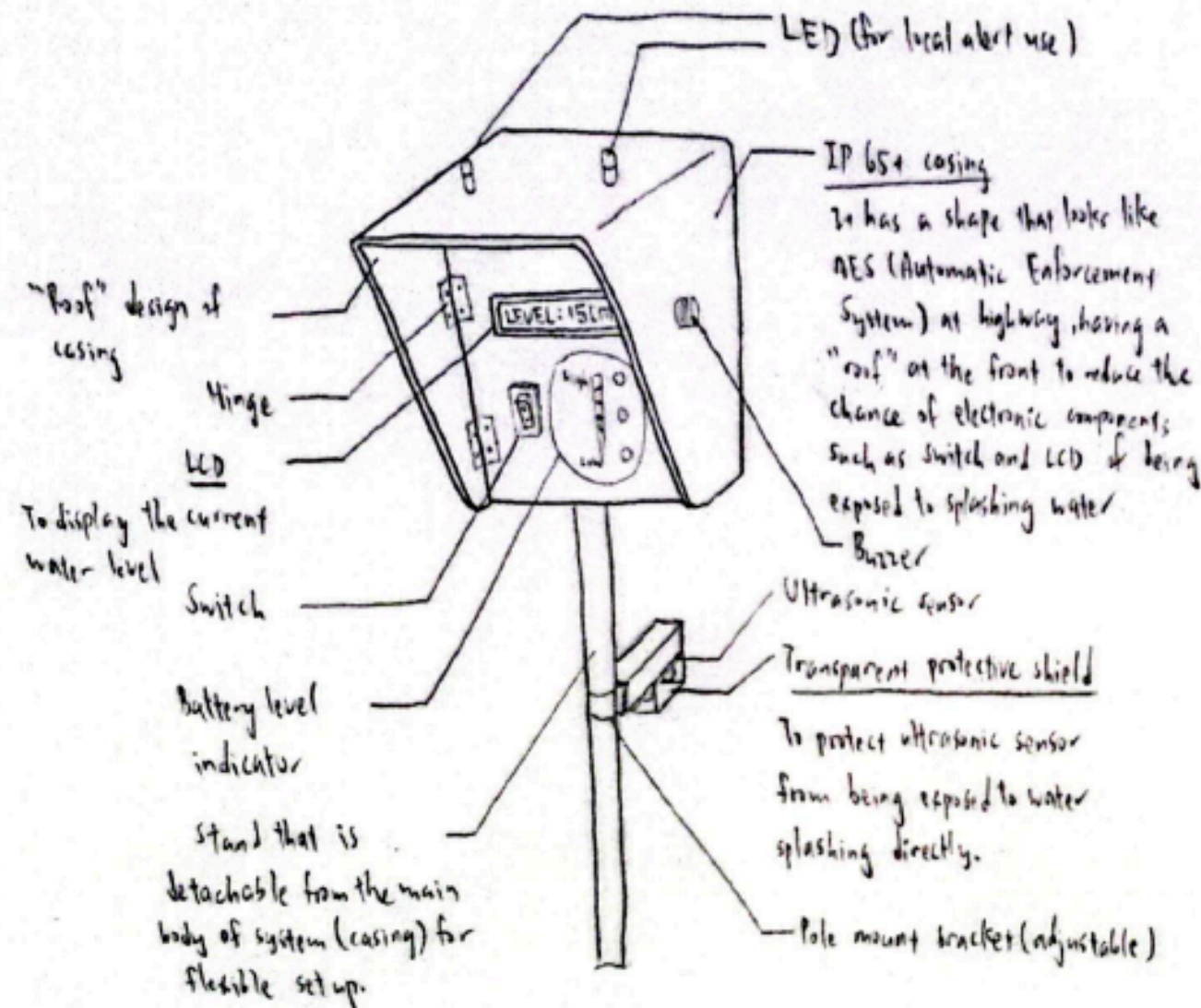
- ① Detect water level
- ultrasonic sensor
- ② Process & analyse data
- avoid false triggering
- ③ Communicate alerts
- GSM message to registered phone
- ④ Early warning
- audio alert
- ⑤ Solar powered supply
- ⑥ Enclosed with vent drain
- ⑦ Text message based
- ⑧ mounted at water
- ⑨ Seasonal use

Concept Redevelopment

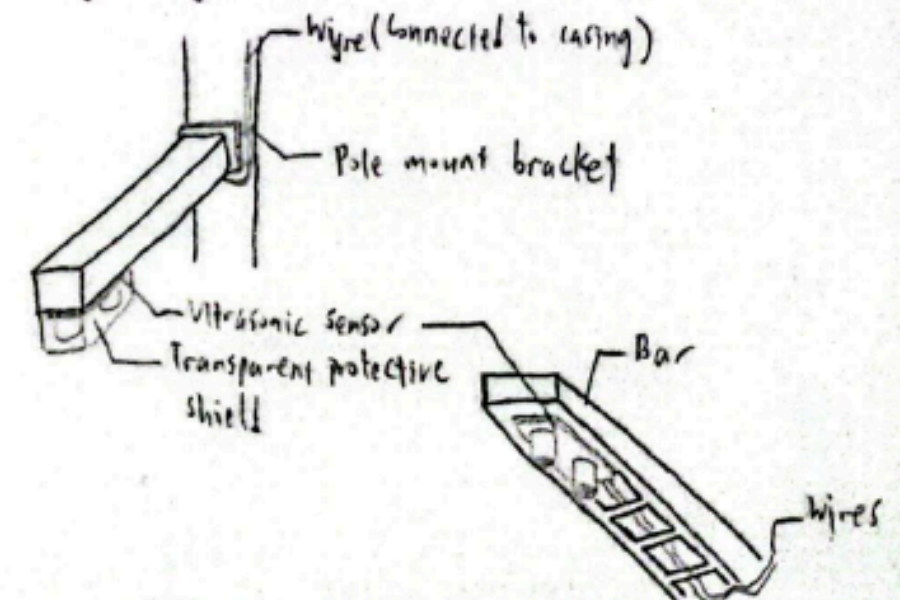
Conceptual Redevelopment

NAME: HEW YEE HANG

GROUP: 7

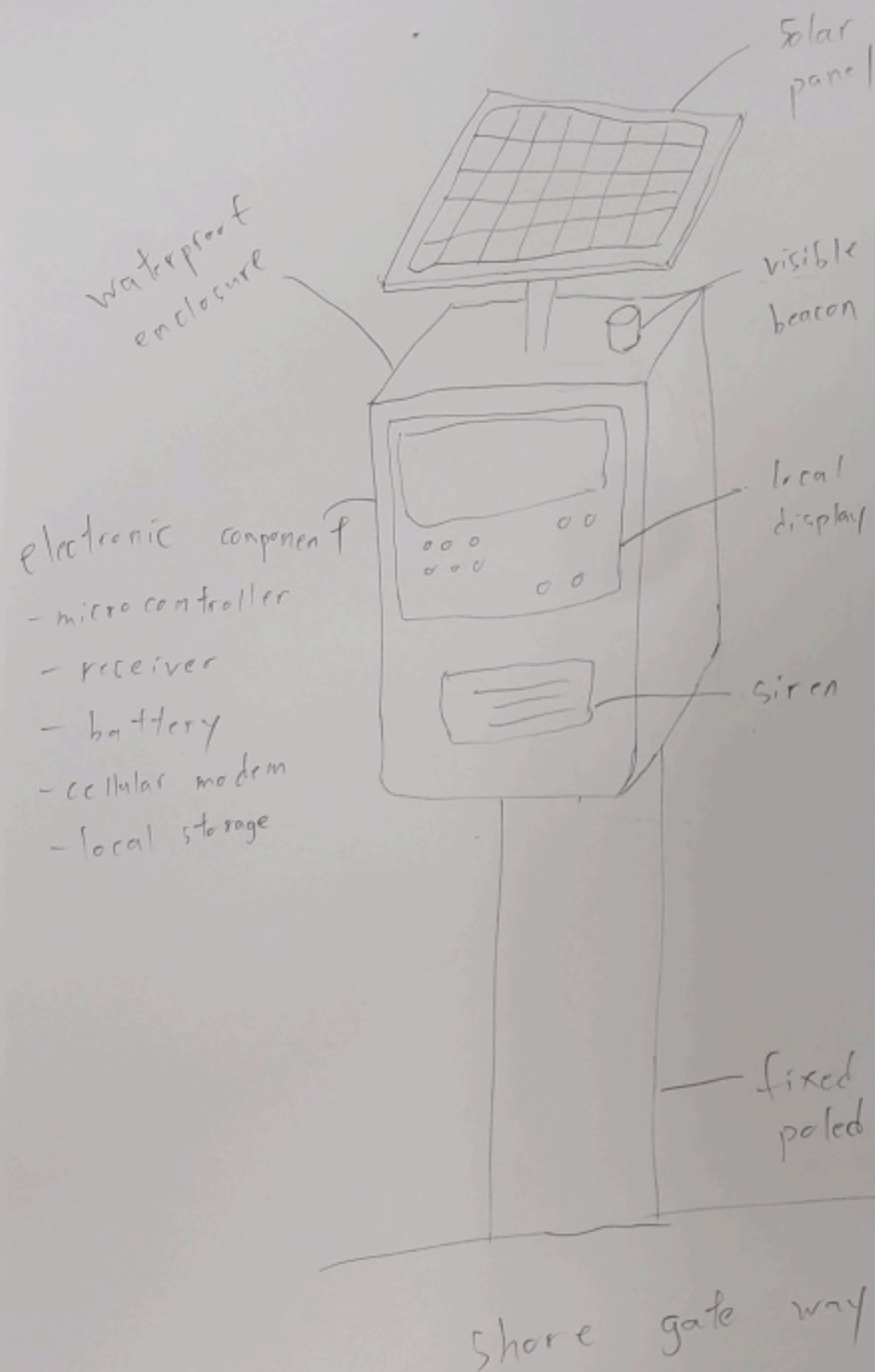


(The wiring in the casing will be fixed on the wall of casing neatly for easier maintenance in the future)



*The bar (as shown in figure above) is hollow with the design at the bottom. This design allows us to insert and hide the wires easily into the bar and it can protect the wires from being exposed to water splashing too.

Hew Yee Hang

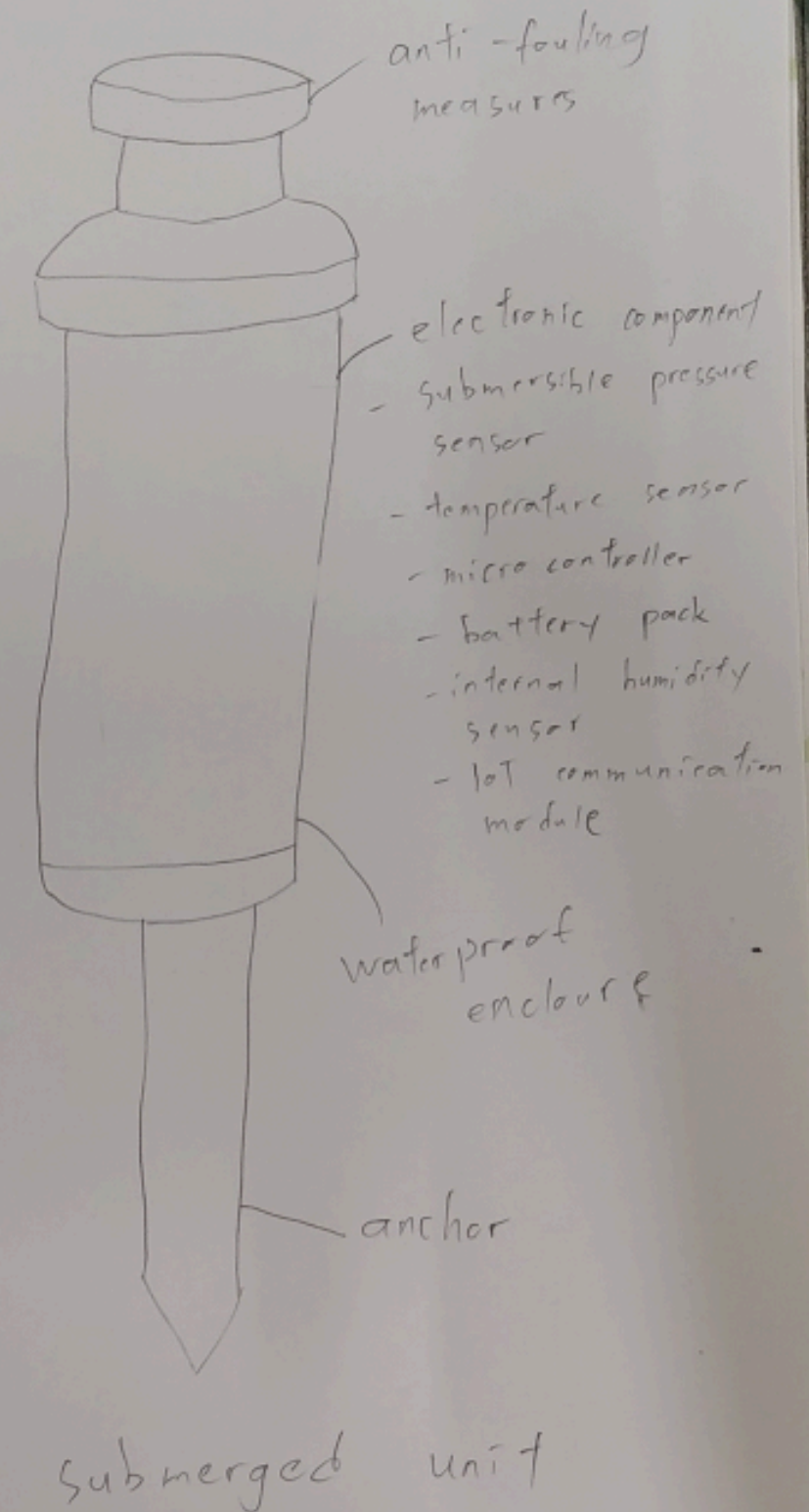


Concept 4:

A fully submerged pressure-sensing unit that senses water depth and reports to a land receiver

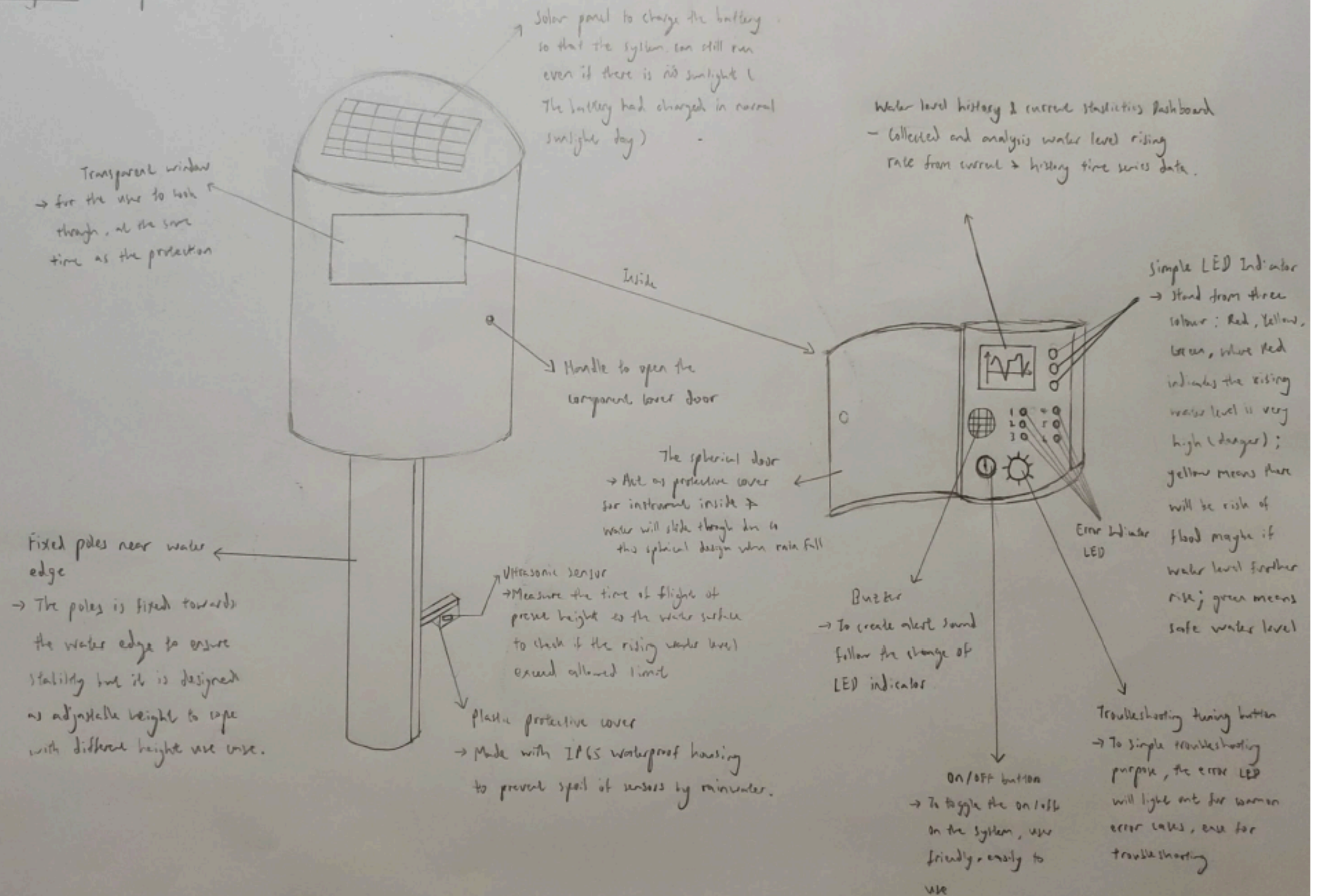
Explanation:

The pressure sensor measures the water level by sensing how much water weight is pressing down on it and converting that pressure into an accurate depth reading.



Lawrance Sim Lip We

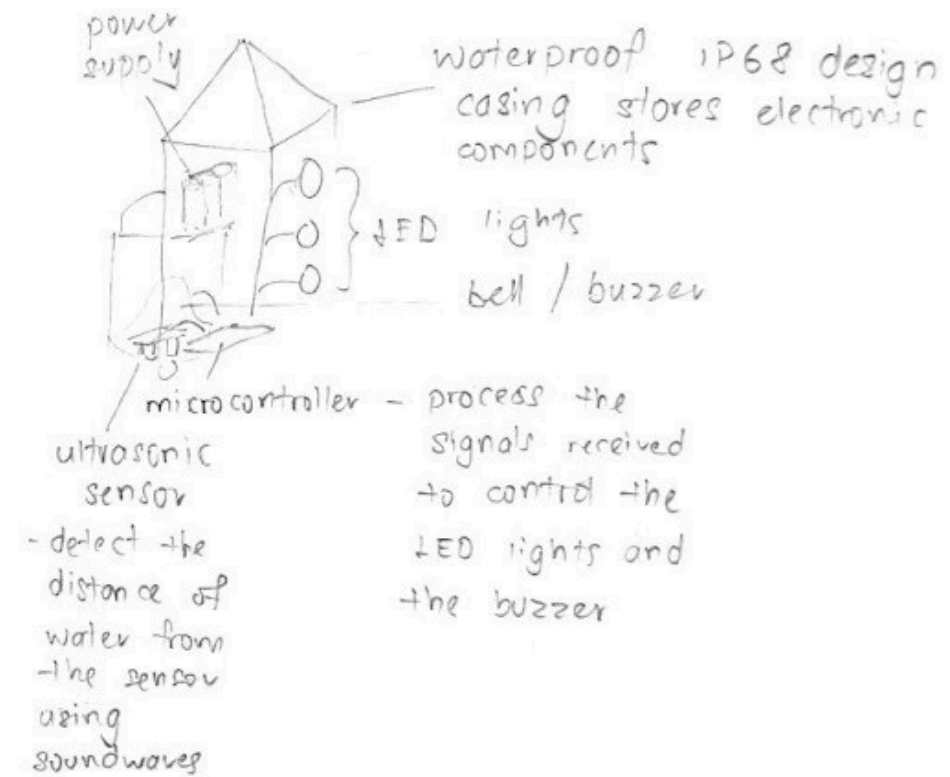
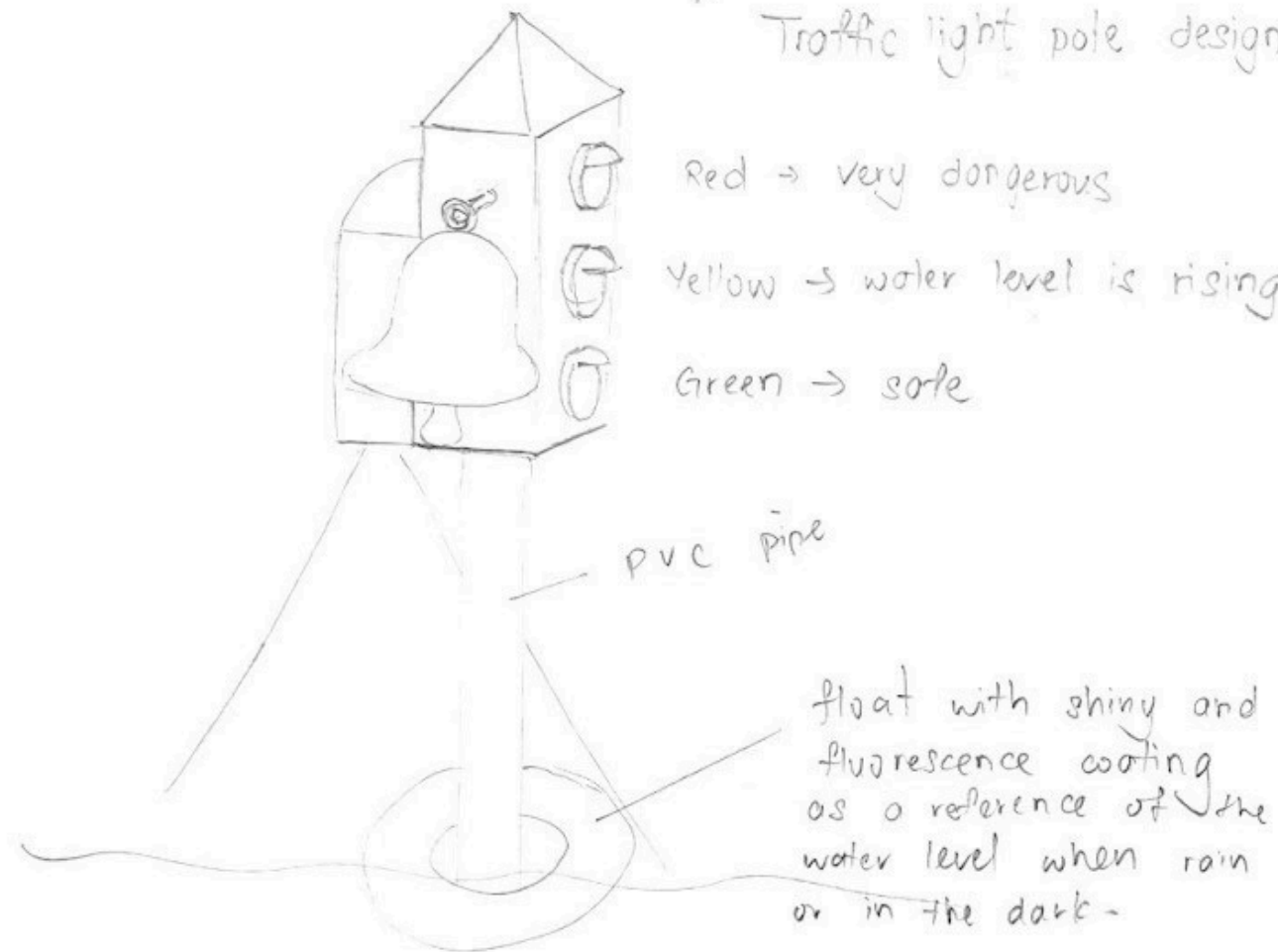
Conceptual Redevelopment (Group 7)



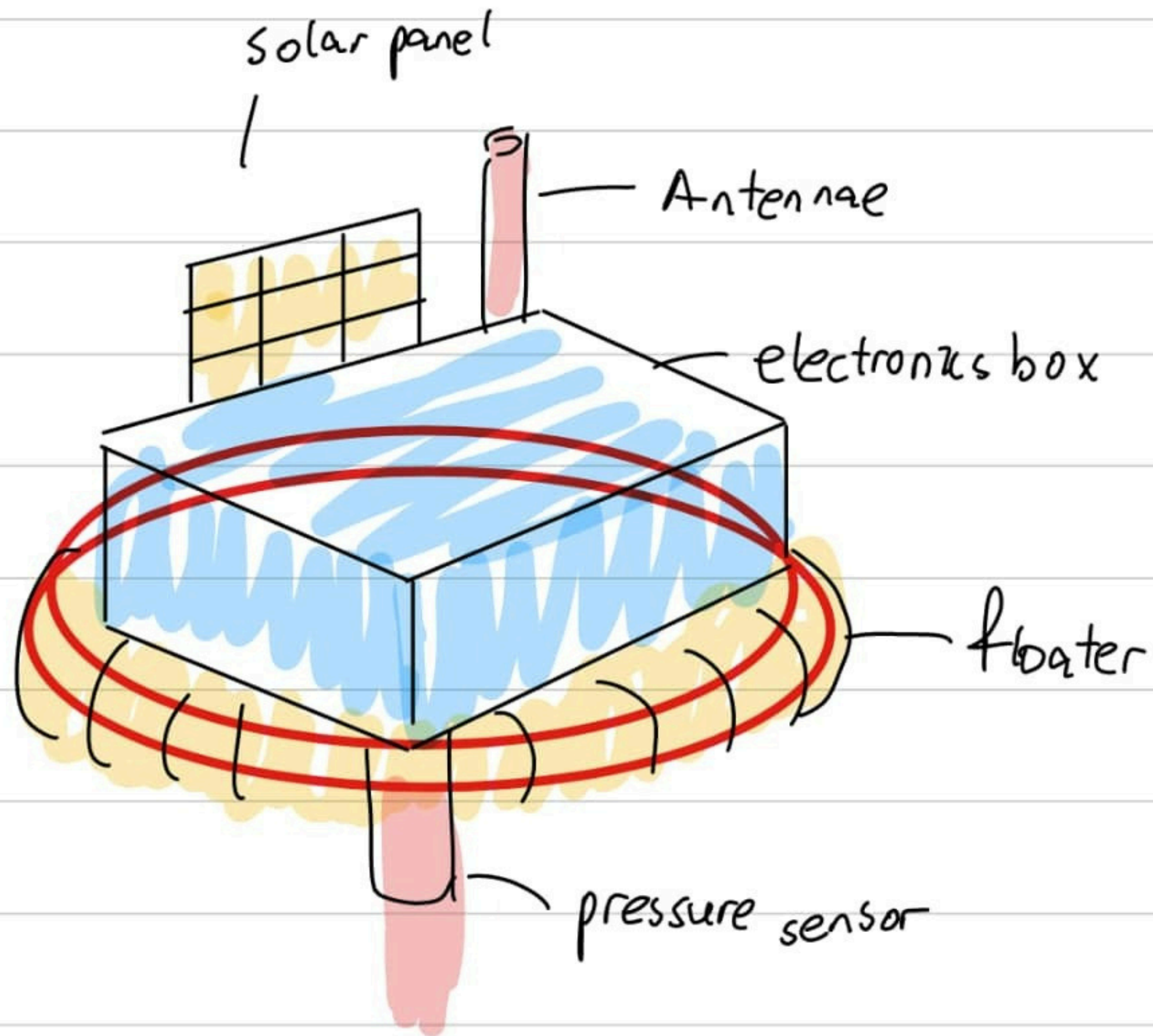
Lawrance Sim Lip We

Liew Shan Jie A22EM0107
Redeveloped Conceptual Design

"Traffic light pole design flood detection system"



Liew Shan Jie



Pang Zi Hao

Assessment Type: Group

Section 5

Concept Selection: Screening , and Scoring matrix

Note: attach your snapshot images

Screening matrix

Note:

- S1=Conceptual sketching 1
- S2=Conceptual sketching 2
- S3=Conceptual sketching 3

Designer	Hew Yee Hang			Lawrance Sim Lip We			Kelvin Lim Dong Xhian			Pang Zi Hao			Liew Shan Jie		
Design Criteria	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
1. Water level detection accuracy	+	0	-	0	-	0	-	+	-	-	-	+	0	-	-
2. Water level detection range	+	0	+	+	-	+	0	+	+	+	+	+	0	+	-
3. Response time	+	0	-	+	-	+	-	+	0	-	-	+	0	-	-
4. Weather (water) resistance	-	+	+	+	+	-	+	+	+	-	-	0	0	+	+
5. Reliability and low false alarm rate	0	0	-	-	-	-	0	+	-	0	-	+	0	+	-
6. Sound intensity of audible alarm	+	+	-	0	-	0	+	0	0	-	+	-	0	-	-
7. Visible alert system performance	0	+	-	+	-	-	+	0	-	+	+	-	0	-	-

8. Overall local alert system performance	0	-	+	+	-	+	+	0	+	+	0	+	0	+	-
9. Endurance	0	+	+	-	-	+	+	-	0	-	0	-	0	-	0
10. Weight	-	+	-	-	+	+	-	+	+	+	+	-	0	-	+
11. Overall size	-	+	-	0	-	+	+	+	+	+	+	-	0	-	+
12. Usability rating (ease of use)	0	-	+	+	+	+	+	0	+	+	0	0	0	+	0
13. Ease of installation	-	+	+	0	+	-	-	-	+	-	+	+	0	-	-
14. Ease of maintenance	+	+	+	-	+	-	0	+	-	+	0	0	0	-	+
15. Cost & manufacturability	0	+	0	-	-	-	-	-	-	-	+	-	0	-	+
16. Ease of assemble	+	0	+	-	+	-	-	-	+	0	+	0	0	0	+
17. Dashboard / UI support	0	-	0	+	-	+	0	+	0	-	0	+	0	+	0
Total Score	2	6	1	1	-5	1	1	5	3	-1	4	1	0	-4	-2

Scoring Matrix

Note:

- S1=Conceptual sketching 1
- S2=Conceptual sketching 2
- S3=Conceptual sketching 3
- CR=Concept redevelopment
- R = Rate
 - 5=Much better than datum
 - 4=Better than datum
 - 3=Equal to datum
 - 2=Worse than datum
 - 1=Much worse than datum
- W= Weightage

[illegible]

1	Water level detection accuracy	0.15	4	0.6	2	0.3	5	0.75	3	0.45	5	0.75	4	0.6	5	0.75	3	0.45	4	0.6	4	0.6
2	Reliability & low false alarm rate	0.10	4	0.4	3	0.3	4	0.4	3	0.3	4	0.4	4	0.4	4	0.4	3	0.3	3	0.3	3	0.3
3	Response time	0.08	4	0.32	3	0.24	4	0.32	2	0.16	4	0.32	3	0.24	4	0.32	3	0.24	3	0.24	4	0.32
4	Weather (water) resistance	0.08	4	0.32	3	0.24	5	0.4	3	0.24	4	0.32	3	0.24	4	0.32	3	0.24	4	0.32	3	0.24
5	Endurance (Power stability)	0.07	2	0.14	2	0.14	3	0.21	2	0.14	4	0.28	3	0.21	5	0.35	2	0.14	4	0.28	5	0.35
6	Water level detection range	0.06	4	0.24	2	0.12	4	0.24	2	0.12	4	0.24	4	0.24	5	0.3	3	0.18	4	0.24	5	0.3

7	Ease of installation	0.05	3	0.15	3	0.15	5	0.25	4	0.2	4	0.2	5	0.25	2	0.1	2	0.1	3	0.15	5	0.25
8	Sound intensity of audible alarm	0.05	4	0.2	3	0.15	4	0.2	3	0.15	3	0.15	3	0.15	3	0.15	4	0.2	4	0.2	1	0.05
9	Cost & manufacturability	0.05	3	0.15	4	0.2	5	0.25	4	0.2	4	0.2	3	0.15	3	0.15	3	0.15	2	0.1	3	0.15
10	Overall local alert (LCD performance)	0.05	4	0.2	3	0.15	5	0.25	2	0.1	4	0.2	1	0.05	4	0.2	5	0.25	4	0.2	1	0.05
11	Dashboard / UI support	0.05	1	0.05	1	0.05	3	0.15	1	0.05	3	0.15	5	0.25	5	0.25	2	0.1	4	0.2	4	0.2

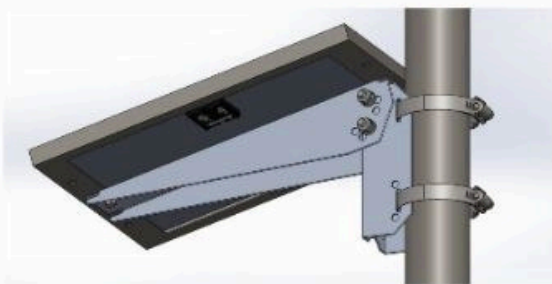
12	Ease of maintenance	0.04	4	0.16	3	0.12	4	0.16	2	0.08	4	0.16	2	0.08	2	0.08	3	0.12	3	0.12	2	0.08
13	Usability rating (ease of use)	0.04	3	0.12	3	0.12	3	0.12	3	0.12	4	0.16	4	0.16	4	0.16	4	0.16	3	0.12	3	0.12
14	Visible alert system performance	0.04	4	0.16	4	0.16	3	0.12	4	0.16	3	0.12	1	0.04	3	0.12	5	0.2	4	0.16	1	0.04
15	Ease of assembly	0.03	3	0.09	3	0.09	3	0.09	4	0.12	4	0.12	5	0.15	2	0.06	3	0.09	2	0.06	4	0.12
16	Overall size	0.03	3	0.09	3	0.09	4	0.12	4	0.12	4	0.12	5	0.15	3	0.09	2	0.06	2	0.06	4	0.12
17	Weight	0.03	3	0.09	3	0.09	3	0.09	4	0.12	4	0.12	5	0.15	3	0.09	2	0.06	2	0.06	4	0.12
Total Score		1.00	57	3.48	48	2.71	67	4.12	50	2.83	66	4.01	60	3.51	61	3.89	52	3.04	55	3.41	56	3.41

Assessment Type: Group

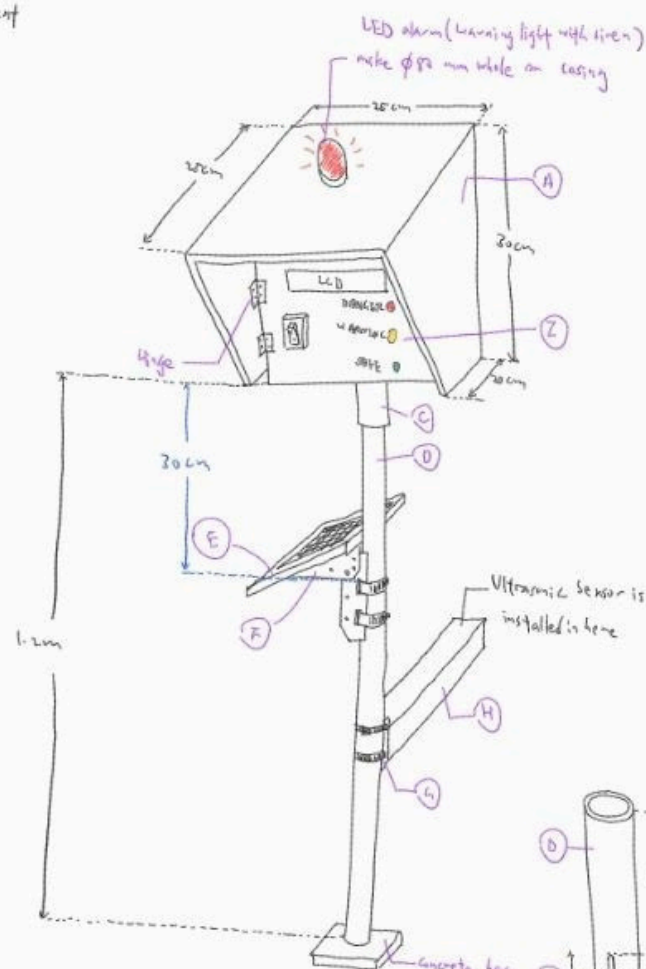
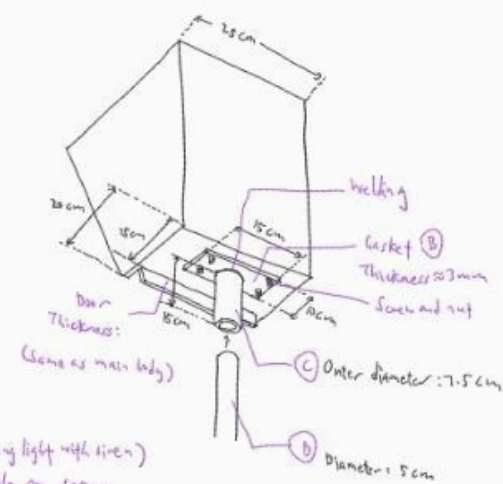
Section 6

Final Design

Note: attach your snapshot images

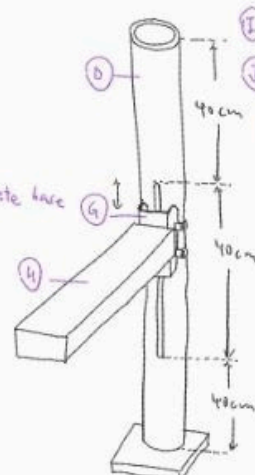


Solar panel attachment

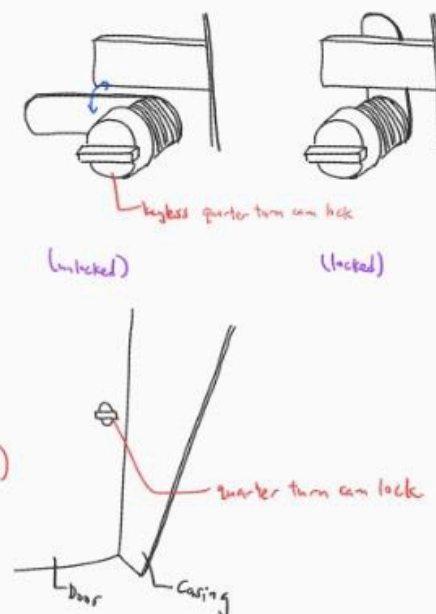
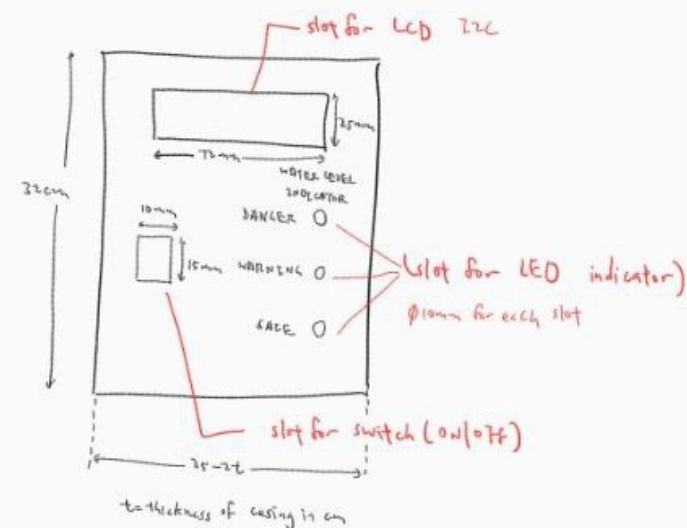


- (A) Casing
- (B) Gasket
- (C) Connector between gasket and pole
- (D) Pole
- (E) Solar panel
- (F) Pole-mount bracket for solar panel
- (G) Pole-mount bracket for extended structure that holds sensor
- (H) Extended structure where sensor is mounted inside.
- (I) Door
- (J) Detachable platform for placing electronic circuit

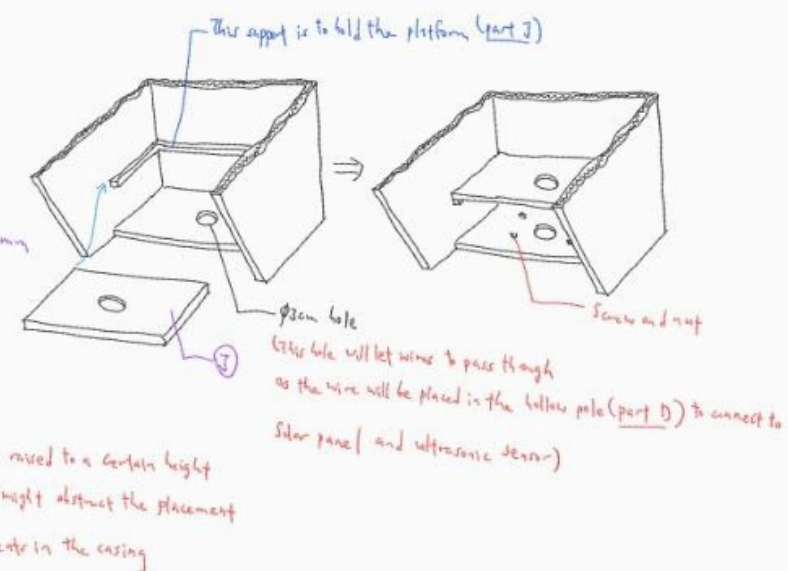
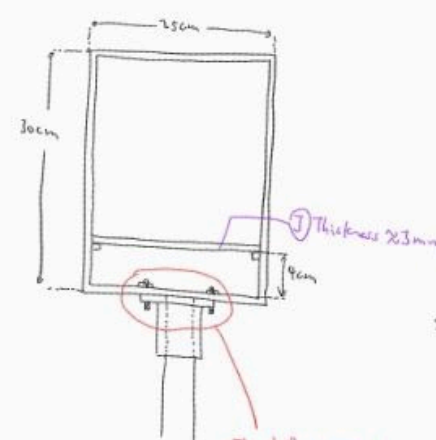
Provide total 40cm adjustable length



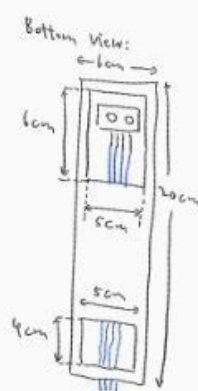
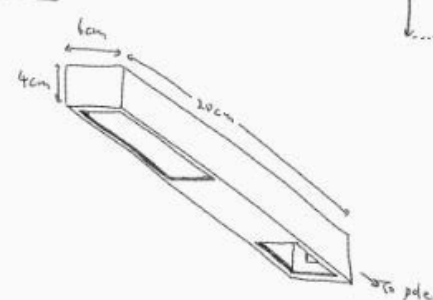
PART I



Front view of the casing

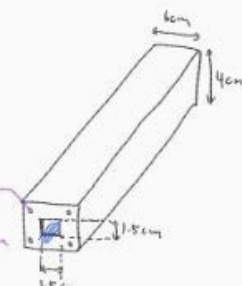


PART II



Holes for screw & nut to let it attach on bracket

Thickness 2mm (choose nearest thickness in solidwork)



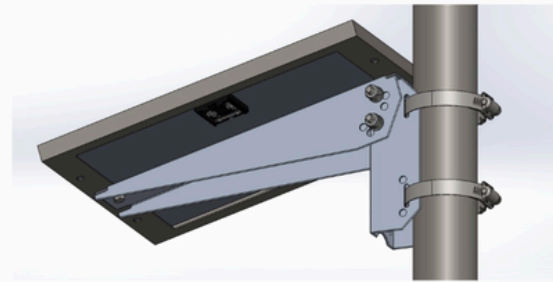
Assessment Type: Individual

Section 7

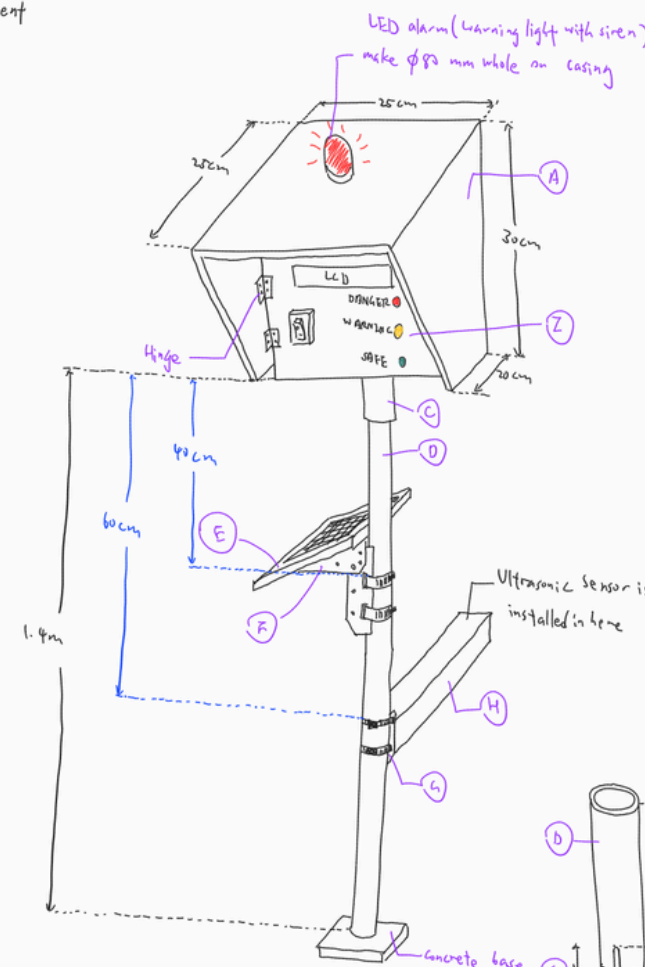
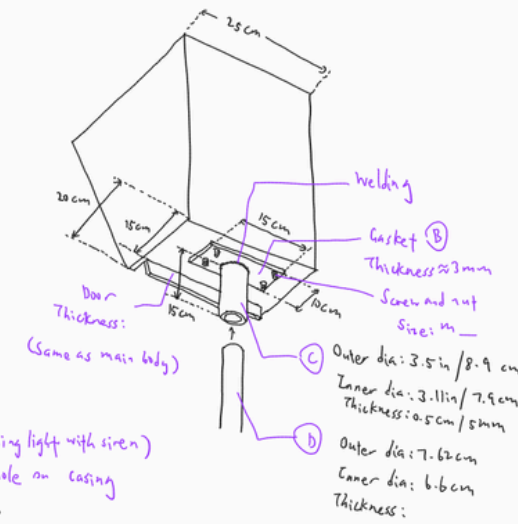
Embodiment Design & Detail Design Sketches

Each group member must submit their individual contribution for this section. Please ensure your name is clearly written on all submitted materials.

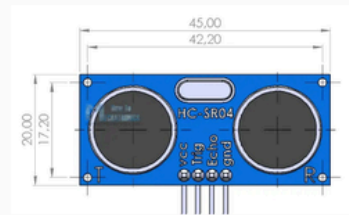
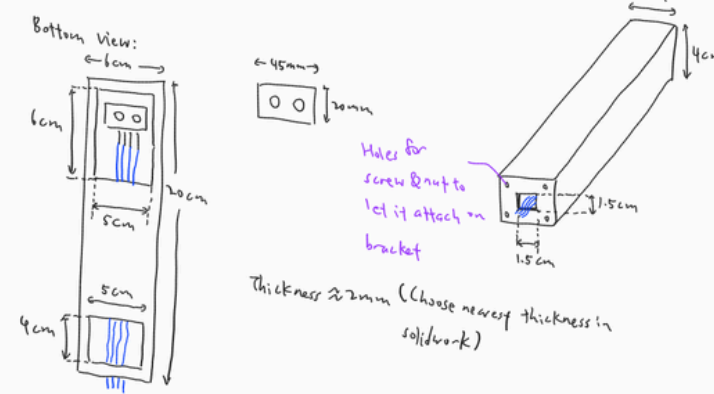
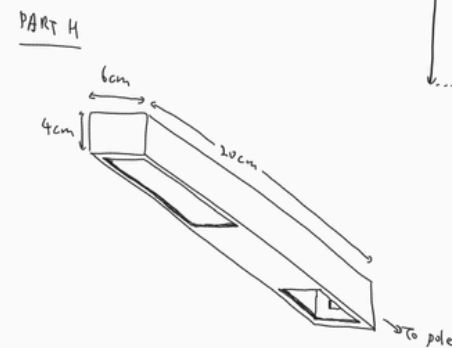
Note: attach your snapshot images



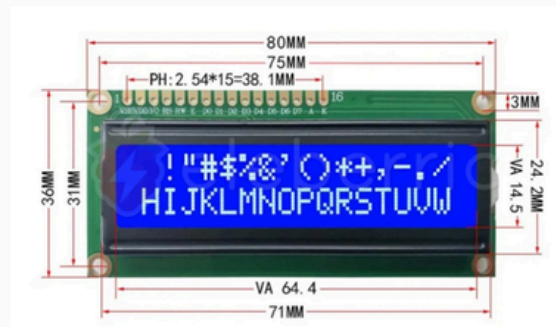
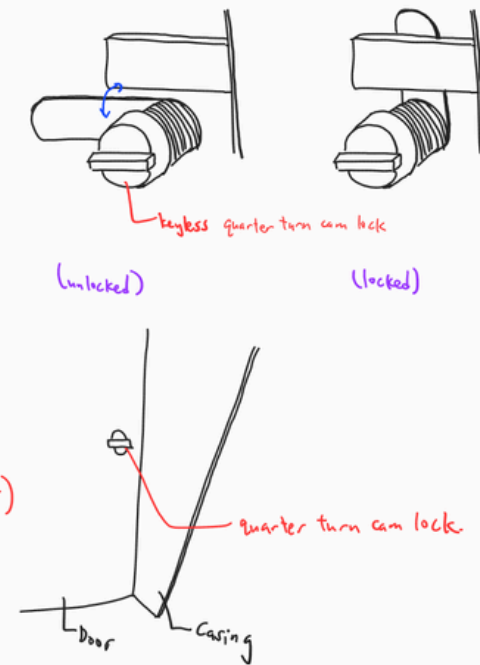
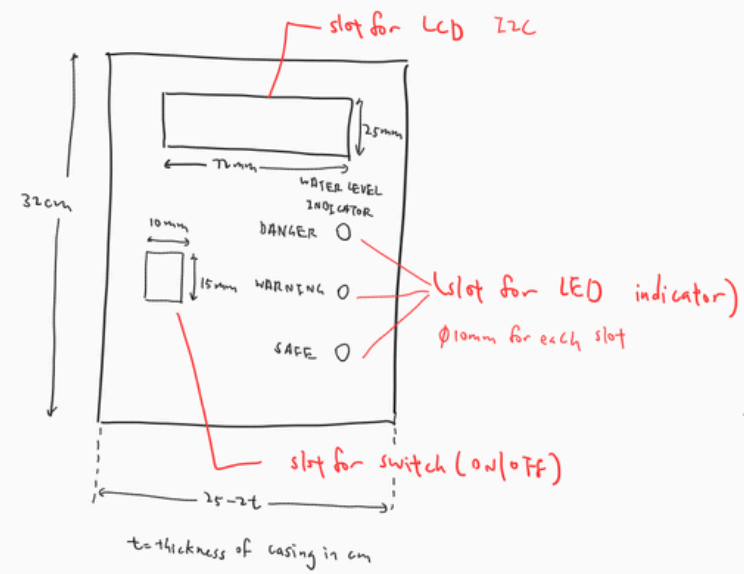
Solar panel attachment



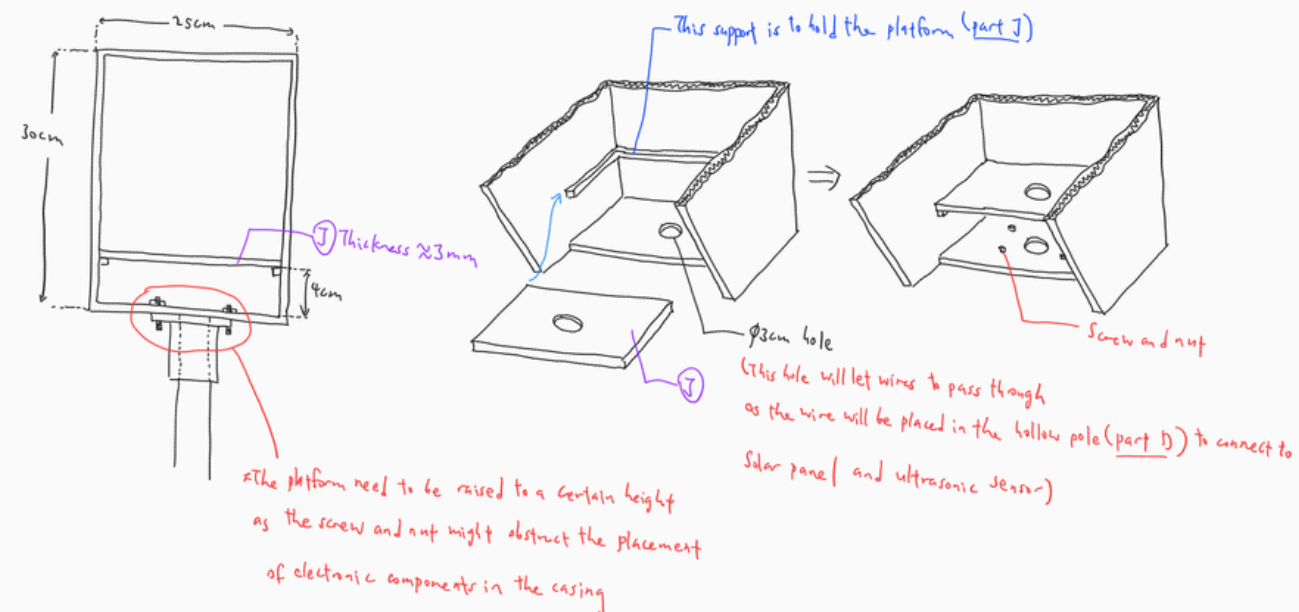
- (A) Casing
Options: 3D printing
Options: wood
 - (B) Gasket—Need in prototype
 - (C) Connector between gasket and pole (PVC)
 - (D) Pole (PVC)
 - (E) Solar panel
 - (F) Pole mount bracket for solar panel
 - (G) Pole mount bracket for extended structure that holds sensor
 - (H) Extended structure where sensor is mounted inside.
 - (I) Door
 - (J) Detachable platform for placing electronic circuit
- * Provide total 40 cm adjustable length

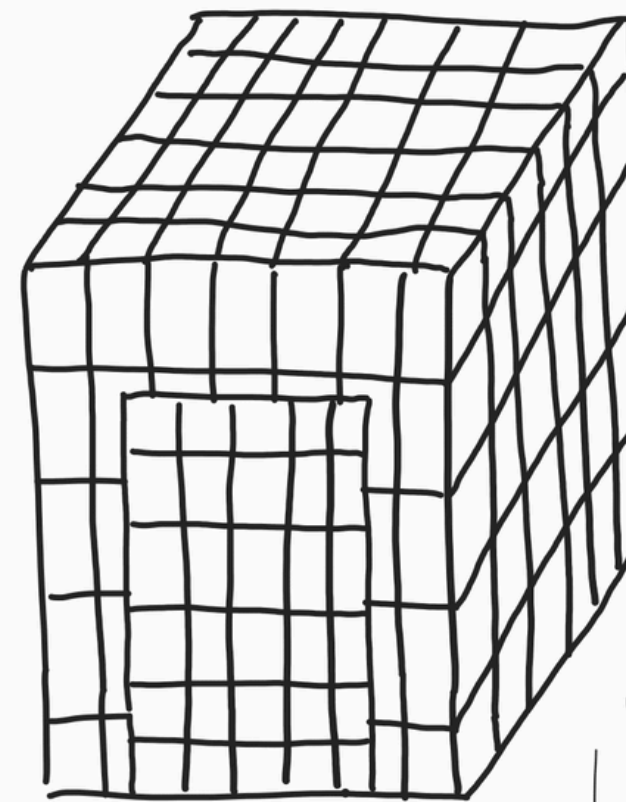


PART I

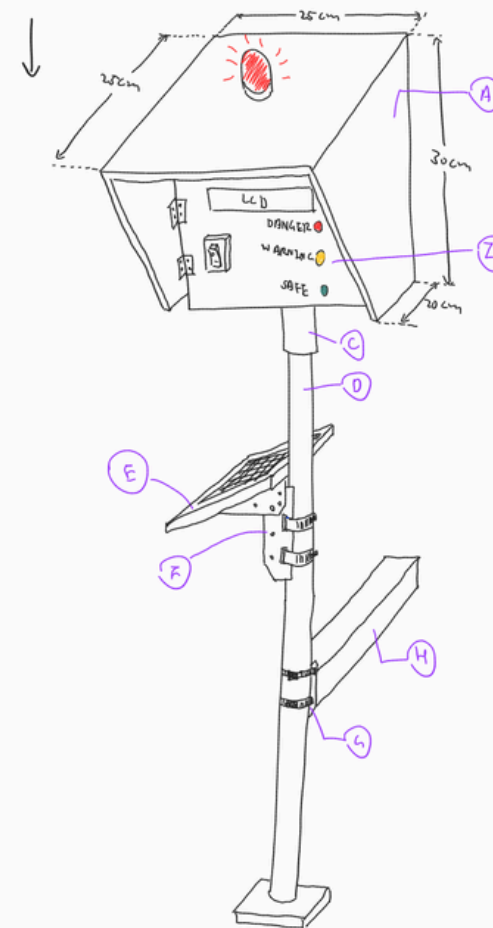


Front view of the casing





Cage: To protect the system from animals and people



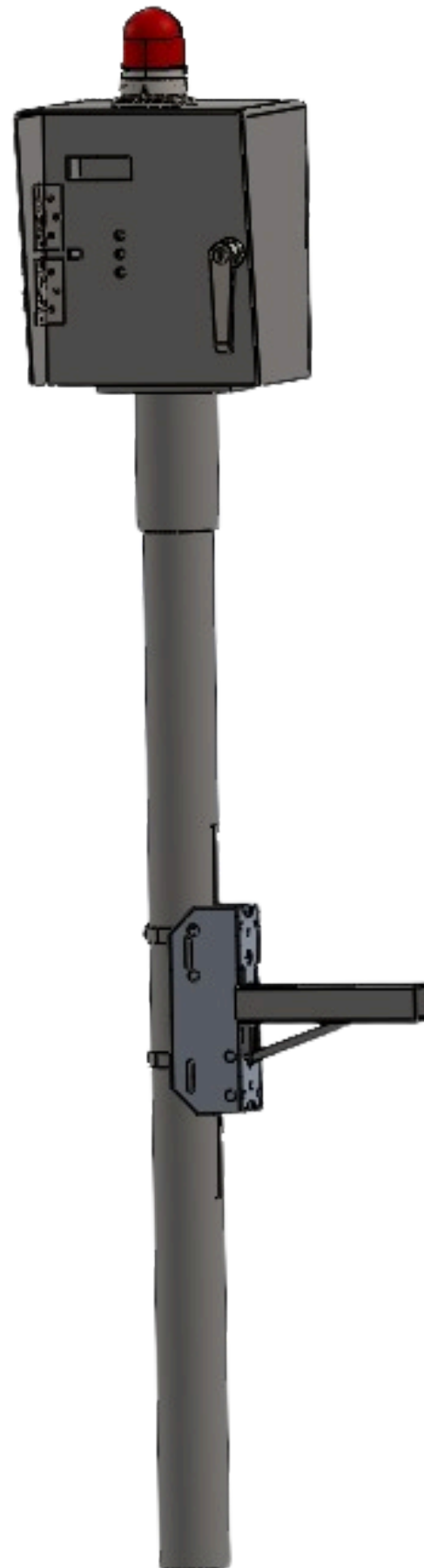
Assessment Type: Group / Individual

Section 8

3D CAD Modeling/Engineering Drawing

Each group member must submit their individual contribution for this section. Please ensure your name is clearly written on all submitted materials.

Note: attach your snapshot images



ITEM NO.	PART NUMBER	QTY.	Part Cost
1	Casing	1	
2	Rod1	1	
3	Rodholder	1	
4	Worm clamp	2	
5	Door	1	
6	Doorhandle	1	
7	Hinge Assembly	2	
8	Siren	1	
9	M5 Screw	16	
10	Pole Bracket	1	
11	Electronics casing	1	

Assessment Type: Group

Section 9

Result: Prototyping

Note: attach your snapshot images



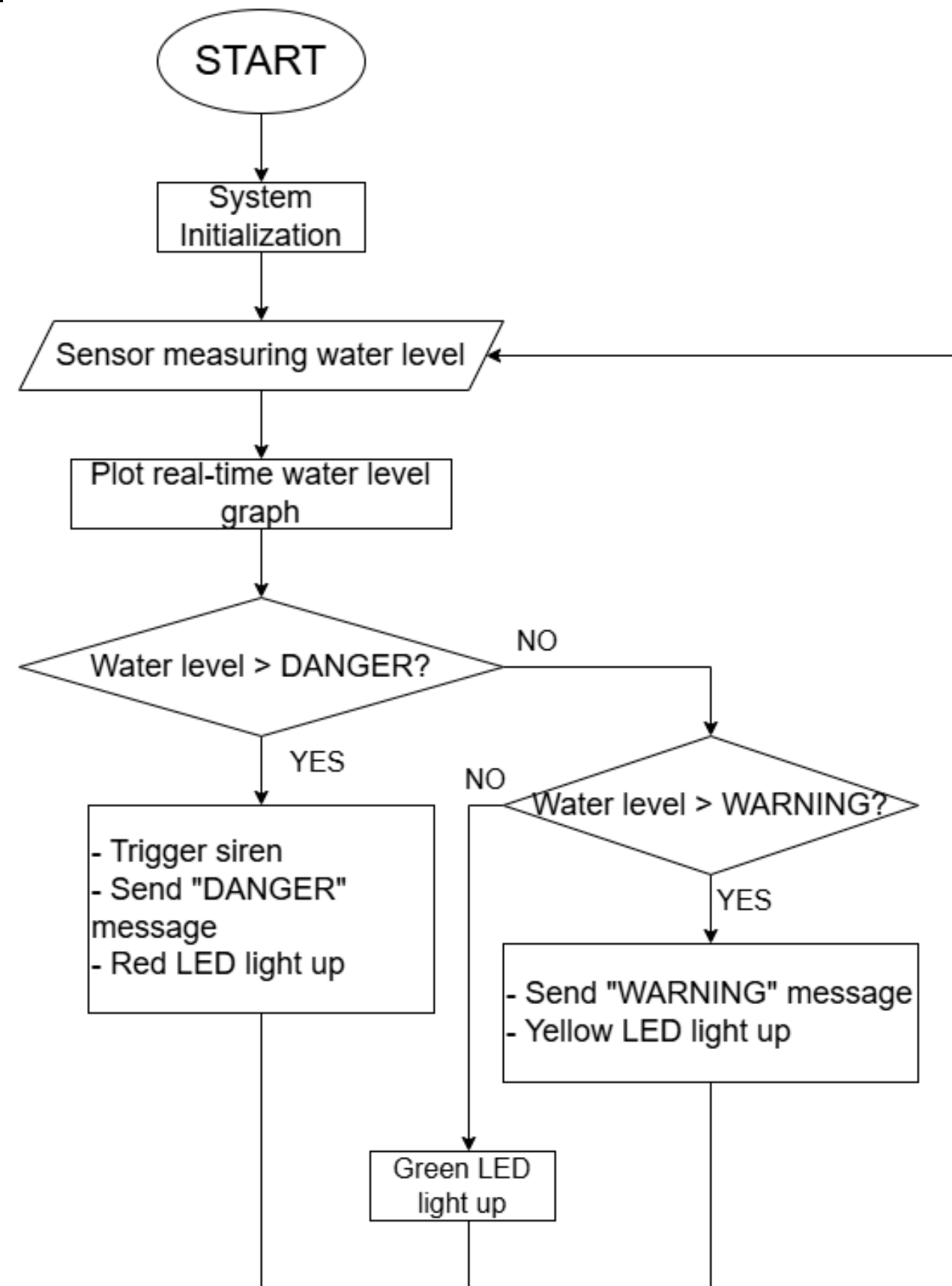
Assessment Type: Group

Section 10

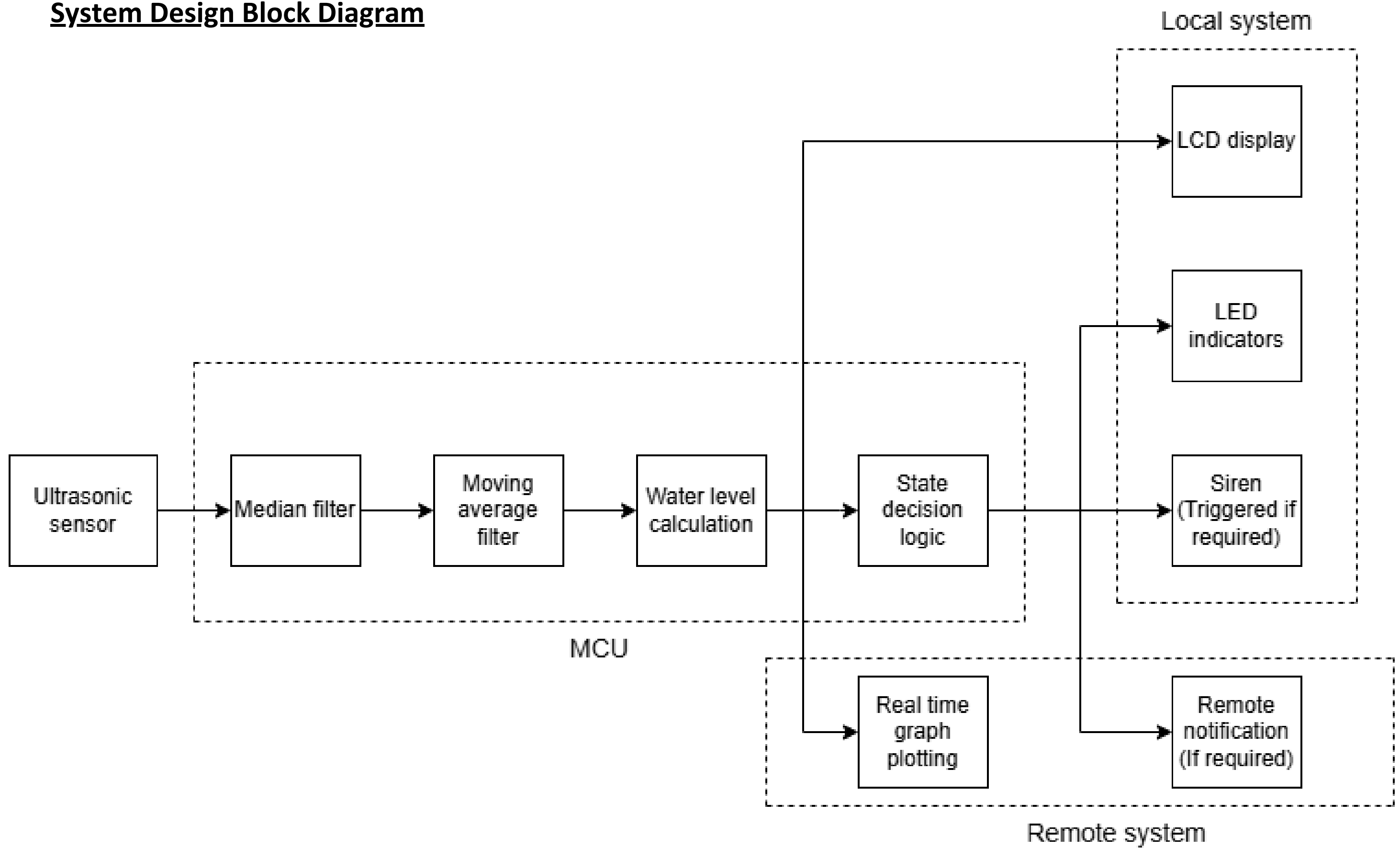
Results: Proof of Function/Design Concept

Note: attach your snapshot images to shows how it works

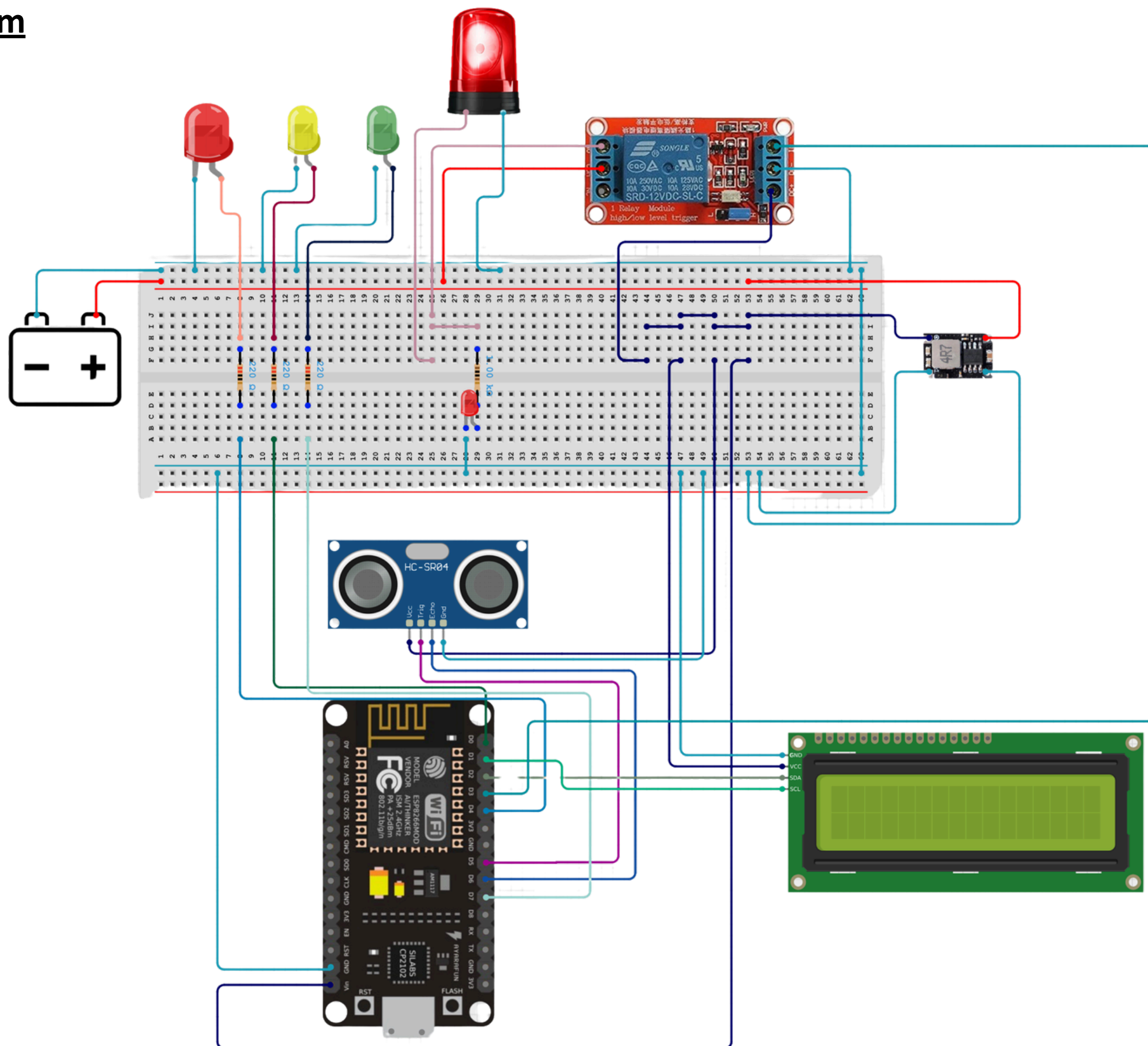
System Design Flowchart



System Design Block Diagram



Circuit Diagram



Video Link

<https://youtu.be/sLGrr76ABu4>